AI Online Speech Disorder Screening and Therapy Assistant

R26-IT-086

Component - Noise Robust Online Speech Screening and Self Supervised Feature Extraction

Project Proposal Report

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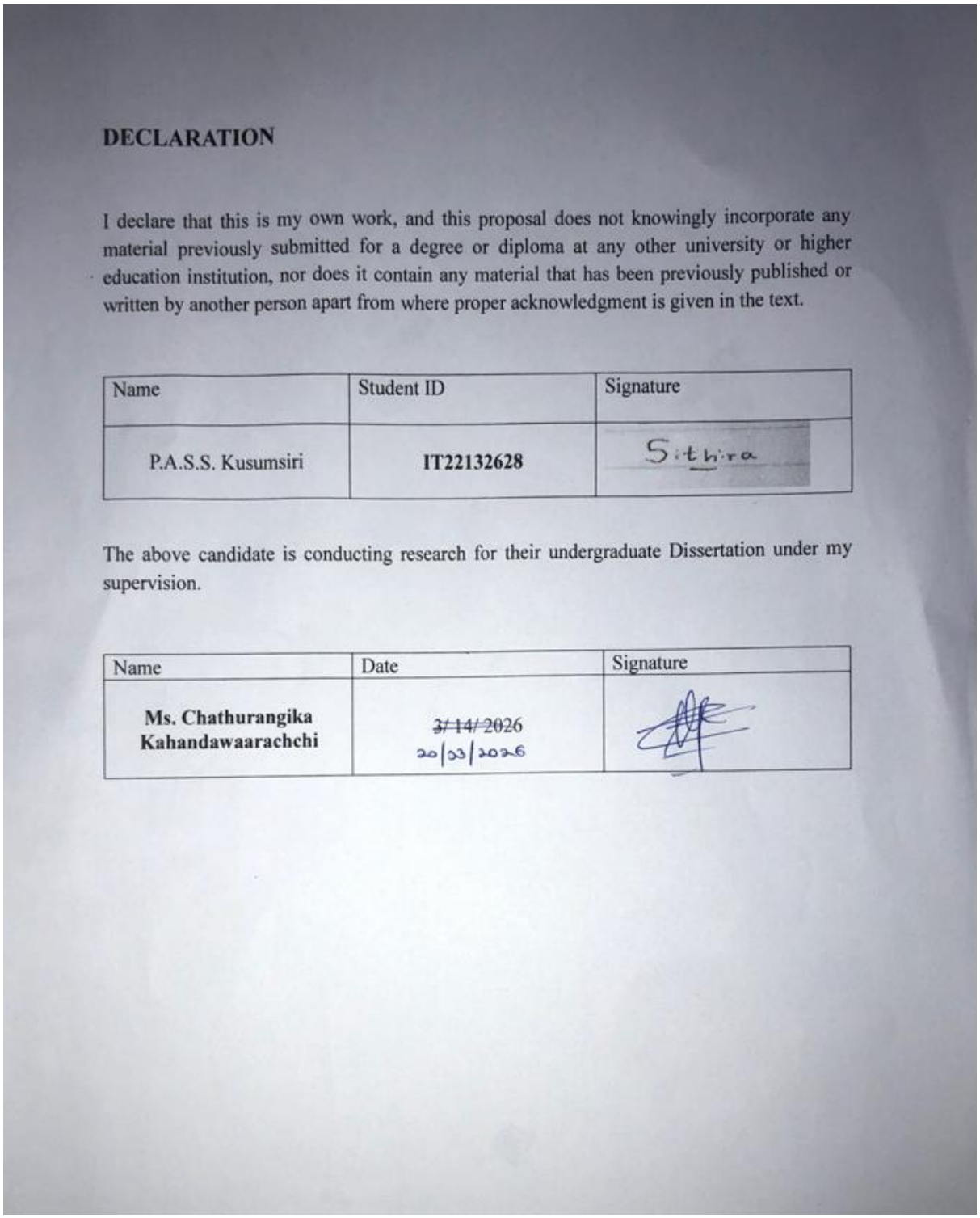
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Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION



DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Speech disorders - including stuttering, dysarthria, and articulation disorders affect approximately 5 - 8% of the global population, yet access to formal speech-language pathology (SLP) assessment remains constrained by geographical, financial, and systemic resource barriers [1]. Automated online speech screening presents a scalable pathway toward early detection. However, most existing systems are designed for controlled, laboratory-quality acoustic environments and rely on standard automatic speech recognition (ASR) pipelines that degrade significantly when applied to pathological speech in real-world noise conditions.

This research proposes the design and implementation of a Noise Robust Online Speech Screening and Self Supervised Feature Extraction component, forming the front-end data acquisition and feature layer of a broader multi-component AI speech disorder assessment system (R26-IT-086).The core technical contribution addresses a critical and underexplored failure mode - standard ASR transcription is systematically unreliable on disordered speech, causing fluency metrics derived from transcription words per minute (WPM), disfluency event density, articulation rate to be materially distorted. This research introduces an ASR uncertainty aware feature extraction pipeline that explicitly models ASR confidence and weights fluency metrics, accordingly, combined with an adaptive speech quality gating mechanism to suppress unreliable measurements before they propagate downstream.

Pre-trained speech foundation models (HuBERT, wav2vec 2.0, Whisper) are fine-tuned on disorder-specific English corpora (TORGO, SEP-28k, UASPEECH) to yield disorder discriminative acoustic representations without dependence on accurate transcription.
A structured three task elicitation protocol (reading passage, picture description, spontaneous speech) captures multiple speech production modalities in a single online session, enabling richer disorder characterization than single task paradigms. Component robustness is validated through cross-corpus experiments across varying SNR levels, microphone types, and noise categories.
Expected outcomes include a clinically reliable, noise-validated fluency feature set, a fine-tuned pathological speech encoder, and a benchmarked robustness report. This work directly supports SDG 3 (Good Health and Well-being) and SDG 10 (Reduced Inequalities) by enabling accessible, technology-mediated early speech disorder screening.

Keywords: speech disorder screening, self-supervised learning, ASR uncertainty estimation, noise-robust feature extraction, fluency analysis.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ASR	Automatic Speech Recognition
CTC	Connectionist Temporal Classification
DAF	Delayed Auditory Feedback
DL	Deep Learning
DSR	Design Science Research
GPU	Graphics Processing Unit
HuBERT	Hidden Unit Bidirectional Encoder Representations from Transformers
ML	Machine Learning
MUSAN	Music, Speech, and Noise corpus
NFR	Non-Functional Requirement
ONNX	Open Neural Network Exchange
PCM	Pulse Code Modulation
SDG	Sustainable Development Goal
SLP	Speech-Language Pathologist / Pathology
SNR	Signal-to-Noise Ratio
SUS	System Usability Scale
TLS	Transport Layer Security
VAD	Voice Activity Detection
WADA	Waveform Amplitude Distribution Analysis
WPM	Words Per Minute

Table 1: List of Abbreviations

LIST OF APPENDICES

1. SEP-28k (https://www.kaggle.com/datasets/ikrbasak/sep-28k) for stuttering event detection
2. TORGO (https://www.kaggle.com/datasets/pranaykoppula/torgo-audio/data)for dysarthria severity modeling
3. UA-Speech (https://www.kaggle.com/datasets/aryashah2k/noise-reduced-uaspeech-dysarthria-dataset) for cross-dataset validation

1. INTRODUCTION

1.1 Background and Context

Real-World Problem

Speech and communication disorders constitute a significant and often under recognized public health challenge. Globally, stuttering alone affects approximately 70 million individuals (roughly 1% of the population), while dysarthria and articulation disorders affect a further 2-5% of adults. many attributable to neurological conditions including stroke, cerebral palsy, Parkinson's disease, and traumatic brain injury [2]. In low- and middle-income countries, including Sri Lanka, SLP workforce density remains critically low. The World Health Organization estimates fewer than one trained SLP per 200,000 people in many South Asian contexts [3]. This structural gap means that individuals with speech disorders frequently go undiagnosed for years, missing critical windows for early intervention during which neuroplasticity driven recovery is most effective.

Beyond clinical access, the logistical barriers of in-person SLP assessment travel costs, appointment wait times of 6-24 months in under-resourced regions, social stigma, and scheduling constraints collectively suppress help-seeking behavior. As a result, severity at first clinical contact is often considerably higher than it would have been with earlier screening.

Why Automated Screening Matters

Automated speech disorder screening offers a paradigm shift rather than replacing clinicians. It serves as a scalable first contact triage layer, enabling population level identification of individuals at risk and routing them to specialist care. An online, browser accessible screening system eliminates geographical and logistical barriers entirely, enabling self-administered assessment from any device with a microphone.

However, realizing this vision introduces a fundamental engineering challenge, real-world audio captured through low-quality microphones, in noisy home or office environments, with the acoustic characteristics of disordered speech, presents conditions that are far removed from the laboratory-recorded corpora on which existing speech processing models have been trained. This mismatch is the central technical problem this research addresses.

Statistical Evidence

Disorder	Global Prevalence	Key Impact
Stuttering	~1% (~70M people)	Persists in adulthood in 25% of cases, employment and social barriers
Dysarthria	2-3% adults	Present in 30% of stroke survivors and 50% of Parkinson's patients
Articulation disorders	8-9% children	Residual effects in ~1% of adults, impacts literacy and employment
SLP to population (South Asia)	1: 200,000	vs. 1:10,000 high income countries (WHO, 2017)

Table 2: Statistical Evidence

Limitations of Existing Solutions

Existing automated screening systems including commercial tools such as Stammurai, SpeechViewer, and research prototypes share three fundamental limitations. First, they evaluate speech recorded in controlled, quiet environments performance on real-world noisy speech has rarely been benchmarked. Second, they depend on standard ASR (e.g., Google Speech to Text, Deep Speech) for fluency metric extraction, without accounting for the well-documented failure of these models on pathological speech inputs where word error rates can exceed 60% [7]. Third, they rely on single task speech elicitation (typically read aloud), which fails to capture the disorder relevant variability that emerges across different speech production modalities (spontaneous speech, picture description).

This research addresses all three limitations through an integrated noise robust pipeline, an ASR-uncertainty-aware metric computation framework, and a multitask elicitation protocol.

SDG and Research Cluster Alignment

- SDG 3 (Good Health and Well-being) - Enabling accessible, early-stage health screening for a population currently excluded from formal SLP assessment.
- SDG 10 (Reduced Inequalities) - Technology-mediated screening reduces socioeconomic and geographic barriers to diagnosis.
- Research Cluster- Intelligent Systems, Human-Computer Interaction, Health Informatics.

1.2 Literature Review

Self-Supervised Learning for Speech Representation

Baevski et al. [8] introduced wav2vec 2.0, a self-supervised framework that teaches powerful speech representations by masking portions of the latent speech signal and solving a contrastive prediction task. The model demonstrated state-of-the-art performance on ASR benchmarks with minimal labelled data, establishing the principle that pretraining on unlabeled speech produces generalizable acoustic features. Critically for this research, wav2vec 2.0 representations capture fine-grained temporal speech structure without requiring transcription labels.

Hsu et al. [9] extended this paradigm with HuBERT (Hidden Unit BERT), which applies offline clustering to assign pseudo-labels to speech frames and trains a BERT style masked prediction model. HuBERT representations have been shown to encode both acoustic and linguistic information at multiple levels of the speech hierarchy, a property valuable for downstream disorder detection tasks. Radford et al. [10] proposed Whisper, a large-scale weakly supervised ASR model trained on 680,000 hours of internet audio. Importantly, Whisper exposes per-token log-probability scores, enabling post hoc estimation of transcription confidence a mechanism this research exploits for ASR uncertainty aware fluency metric weighting.

Disordered Speech Corpora

Rudzicz et al. [11] introduced the TORGO database, comprising 9.7 hours of dysarthric speech from 8 speakers with cerebral palsy and amyotrophic lateral sclerosis (ALS), recorded with simultaneous acoustic and articulatory capture. TORGO remains the most widely used benchmark for dysarthric ASR and severity assessment research. Lea et al. [12] introduced SEP-28k, a 28,000-clip podcast derived dataset of stuttered speech spanning repetitions, prolongations, blocks, and interjections. SEP-28k is notable for its ecological validity, speech is captured in naturalistic, variable acoustic conditions rather than laboratory settings. The UASPEECH corpus [13] provides 103 speakers with dysarthria, segmented into four intelligibility tiers, making it suitable for severity-stratified model training.

Speech Disorder Detection and Severity Assessment

Bayerl et al. [14] demonstrated that fine-tuned wav2vec 2.0 models significantly outperform traditional feature-based methods for stuttering detection, achieving an F1 score of 0.82 on SEP-28k. This result established self-supervised pretraining as the current state of the art foundation for disfluency detection. Vaessen and van Leeuwen [15] further confirmed this by showing that even minimal fine-tuning of wav2vec 2.0 on small stuttering datasets produces strong transfer learning outcomes, suggesting that pre-trained representations generalize well across speakers and recording conditions.

Illa and Ghosh [16] investigated severity assessment for dysarthric speech under low resource conditions, demonstrating that transfer learning from healthy speech pretrained models can bridge the data scarcity problem inherent to pathological speech research. Their work directly motivates this component's strategy of fine-tuning foundation models on the available disordered English corpora.

Noise Robustness and Acoustic Preprocessing

Botelho et al. [17] examined the effect of recording conditions on pathological speech detection, finding that models trained on clean speech degraded significantly at signal to noise ratios (SNRs)

below 15 dB, a threshold routinely crossed in domestic recording environments. This provides empirical justification for the noise robustness evaluation framework proposed in this research. Orozco Arroyave et al. [18] demonstrated that spectral subtraction and Wiener filtering as preprocessing steps improved downstream disorder detection accuracy by 8-12% on noisy recordings, supporting the inclusion of spectral noise suppression in the proposed pipeline.

ASR on Pathological Speech

Multiple studies have documented the catastrophic failure of commercial ASR systems on disordered speech, word error rates exceeding 60-80% have been reported for moderate to severe dysarthria [7]. This has direct implications for fluency metric computation if ASR miscounts words or segments syllables incorrectly, WPM and articulation rate become clinically meaningless. No existing automated screening system addresses this failure mode through confidence weighted metric computation, representing the primary novelty gap this research fills.

Multi-Task Speech Elicitation

Lian et al. [19] showed that combining read and speech tasks with picture description and spontaneous speech tasks significantly improves stuttering detection sensitivity, particularly for individuals whose stuttering is situationally variable. The multi-task elicitation design proposed in this research draws directly from this finding.

1.3 Research Gap

What is Missing

A comprehensive review of existing automated speech screening systems reveals three converging and unaddressed gaps,

1. **ASR reliability on pathological speech is assumed, not modelled.** Every existing fluency measurement pipeline implicitly assumes that ASR transcription is accurate enough to derive reliable word counts, syllable rates, and disfluency segment boundaries. On disordered speech particularly dysarthria and severe stuttering this assumption is violated

at word error rates of 40-80%. Yet no existing system propagates ASR uncertainty into downstream metric computation. This is not merely an engineering oversight. It represents a fundamental validity threat to every fluency score produced by transcription-dependent pipelines.

2. **Noise robustness is evaluated in isolation from disorder detection.** When noise robustness has been studied in speech pathology contexts, it is examined as a separate dimension from disorder detection accuracy. There is no published work that jointly evaluates a speech screening system across both dimensions i.e., how does detection accuracy degrade as a function of SNR for each disorder type? This cross dimensional evaluation is essential for establishing fitness for purpose in real world deployment.

3. **Single-task elicitation protocols undercapacities disorder severity.** Most automated systems use a single speech task, most commonly oral reading. Stuttering and dysarthria severity, however, varies substantially across speech production contexts. A speaker may exhibit minimal disfluency when reading a rehearsed passage but significant disfluency in spontaneous speech. Systems that evaluate only reading speech systematically underestimate severity for a proportion of users, representing a clinical safety concern.

Why Current Approaches Are Insufficient

The limitations above are not addressable by incremental improvements to existing systems. Correcting ASR-based fluency metrics requires architectural changes to the measurement pipeline. Specifically, the integration of token-level confidence scores from ASR models (available in Whisper and wav2vec 2.0 CTC outputs) into a confidence weighted metric computation framework. Similarly, cross corpus noise robustness evaluation requires a structured experimental design that existing tools have not been built to support.

Justification of Novelty

This research contributes to the first published ASR uncertainty aware fluency feature extraction pipeline for automated speech disorder screening. The integration of ASR token level confidence weighting with an adaptive speech quality gate and a disorder-specific fine-tuned foundation model constitutes a novel technical contribution that is neither present nor derivable from any existing published system.

2. OBJECTIVE

2.1 Main Objective

To design, implement, and empirically evaluate a noise-robust online speech screening interface and ASR uncertainty aware self-supervised feature extraction pipeline that produces clinically reliable fluency and acoustic disorder features from unconstrained real-world English speech, validated across multiple acoustic environments and cross-corpus benchmarks.

2.2 Specific Objectives

1. SO1 - Design and implement a three-task online speech elicitation interface (reading passage, picture description, spontaneous speech) that captures multi modal speech production samples via a browser-accessible web platform.
2. SO2 - Develop a multistage audio preprocessing pipeline incorporating adaptive Voice Activity Detection (VAD), spectral noise suppression, and a real-time speech quality assessment gate that filters audio segments falling below a configurable SNR threshold.
3. SO3 - Fine tune pretrained self-supervised speech foundation models (HuBERT and/or Whisper) on disordered English speech corpora (TORGO, SEP-28k, UASPEECH) to generate disorder-sensitive acoustic embeddings without dependence on accurate transcription.
4. SO4 - Design and validate an ASR uncertainty aware fluency metric computation framework in which WPM, articulation rate, pause frequency, and event disfluency density are weighted by per-token ASR confidence scores, reducing the systematic distortion introduced by ASR failure on pathological speech.
5. SO5 - Conduct cross-corpus robustness evaluation experiments across varying SNR levels (0-30 dB), microphone types (condenser, dynamic, smartphone), and background noise categories (office, domestic, outdoor), establishing empirical robustness benchmarks for all extracted feature dimensions.

3. METHODOLOGY

Research Approach

This research adopts a Design Science Research (DSR) methodology, in which the primary artefact the noise-robust speech screening and feature extraction pipeline is iteratively designed, built, and evaluated against prespecified performance criteria. The evaluation follows experimental paradigm-controlled experiments manipulate acoustic conditions (SNR, noise type, microphone) as independent variables and measure feature extraction reliability and downstream disorder detection accuracy as dependent variables.

Technical Methods and Algorithms

Stage 1 - Speech Elicitation Interface A React.js web application is implemented to administer three structured speech tasks in sequence, (i) reading a standardized passage (Rainbow Passage [20]), (ii) describing a standardized picture stimulus (Cookie Theft picture from the Boston Diagnostic Aphasia Examination), and (iii) answering an open-ended prompt for spontaneous speech. Audio is captured at 16 kHz, 16-bit mono via the Web Audio API and streamed to the processing backend over WebSocket. Each task is constrained to 60–120 seconds to standardize elicitation length.

Stage 2 - Preprocessing Pipeline Raw audio enters a three-stage preprocessing pipeline. First, adaptive VAD using Silero-VAD [21] a lightweight neural VAD model segments the audio stream into speech and non-speech intervals, enabling accurate silence and pause duration computation. Second, spectral noise suppression is applied using a Wiener filter estimated from non-speech intervals identified by the VAD stage. Third, a speech quality assessment gate computes the signal-to-noise ratio (SNR) of each speech segment using the WADA-SNR algorithm [22] segments falling below 5 dB SNR flagged for quality warnings rather than discarded, preserving clinical data while marking low-confidence measurements.

Stage 3 - Self-Supervised Feature Extraction Pre-trained HuBERT Large and Whisper Medium models are fine-tuned on the TORGO, SEP-28k, and UASPEECH corpora using a multitask fine tuning objective that jointly optimizes for (a) speaker disorder type classification and (b) severity tier prediction. Fine-tuning is performed with AdamW optimizer (lr = 1e-5, weight decay = 0.01) for 15 epochs with early stopping on validation loss. The resulting encoder generates 768-dimensional disorder-sensitive embeddings per 20ms speech frame, which are aggregated using mean and standard deviation pooling over task level segments.

Stage 4 - ASR-Uncertainty-Aware Fluency Feature Extraction Whisper is used for transcription, with token-level log-probability scores extracted from the decoder output. A per-token confidence weight $w_i = \text{SoftMax}(\text{log_prob}_i)$ is computed. Fluency metrics are then computed as confidence weighted aggregates $\text{WPM_adjusted} = \Sigma(w_i \times \text{word}_i) / T$, where T is the segment duration. Tokens with confidence below a threshold $\theta = 0.3$ are excluded from metric computation rather than contributing noise. Pause duration and frequency are computed directly from VAD segmentation (transcription-independent), preserving reliability under high ASR error conditions.

Stage 5 - Robustness Evaluation A structured cross-corpus evaluation experiment is conducted using recordings from TORGO and UASPEECH, artificially mixed with noise from the MUSAN noise corpus at SNR levels of 0, 5, 10, 15, 20, and 30 dB. Feature stability (measured as coefficient

of variation across SNR conditions) and downstream severity classification accuracy (passed to Component 3) are reported for each noise type and SNR level.

Tools and Platforms

Tool / Framework	Purpose	Justification
Python 3.11, PyTorch 2.x	Core ML pipeline	Industry-standard DL framework with best HuBERT/Whisper integration
HuggingFace Transformers	Model fine-tuning	Pre-trained checkpoints and fine-tuning utilities for HuBERT and Whisper
Silero-VAD	Voice activity detection	Lightweight, real-time capable, ONNX-exportable neural VAD model
Librosa	Audio feature extraction	Robust, well-documented signal processing library for Python
React.js + WebAudio API	Elicitation interface	Browser-native, no installation required; supports live waveform visualization
TORGO, SEP-28k, UASPEECH	Training/evaluation corpora	Gold-standard disordered English speech datasets; ethically approved
MUSAN Noise Corpus	Noise augmentation	Standard noise corpus for cross-SNR robustness evaluation
ONNX Runtime	Model serving	Reduced inference latency, target < 2 s per 60 s audio segment

Table 3: Tools and Platforms

Data Collection and Ethical Considerations

Primary data collection from participants is not required for this component, which trains exclusively on publicly available, ethically approved research corpora (TORGO, SEP-28k,

UASPEECH). All three datasets were collected under institutional ethics approval at their originating institutions. For usability testing of the elicitation interface, a small sample of healthy volunteers (n ≈ 15) will be recruited from within SLIIT with written informed consent. No personally identifiable information is collected. All audio is processed server-side and not retained beyond the session without explicit consent. Full compliance with the Sri Lanka Personal Data Protection Act (2022) is maintained throughout.

Validation Strategy and Performance Metrics

Specific Objective	Metric	Target
Fluency feature reliability (clean speech)	Pearson r vs. manual annotation	**r > 0.85**
Fluency feature reliability (15 dB SNR noise)	Pearson r vs. manual annotation	**r > 0.75**
ASR confidence weighting impact	Reduction in RMSE vs. unweighted baseline	**≥ 15% reduction**
Foundation model fine-tuning	Macro F1 - disorder type classification	**≥ 0.78**
VAD segmentation accuracy	F1 on speech / non-speech classification	**≥ 0.92**
End-to-end processing latency	Time from audio submission to feature output	**< 5 seconds**

Table 4: Validation Strategy and Performance Metrics

4. HIGH-LEVEL SYSTEM ARCHITECTURE

System Overview

The complete R26-IT-086 system comprises four components operating as a sequential processing pipeline. Component 1 (this component) forms the front-end data acquisition and feature extraction layer. Raw audio captured through the web elicitation interface is processed through the

preprocessing pipeline, encoded by the fine-tuned foundation model, and passed as a structured feature vector to Component 2 (Alignment and Articulation Analytics) and Component 3 (Multi-Disorder Severity Classification).

Component Architecture

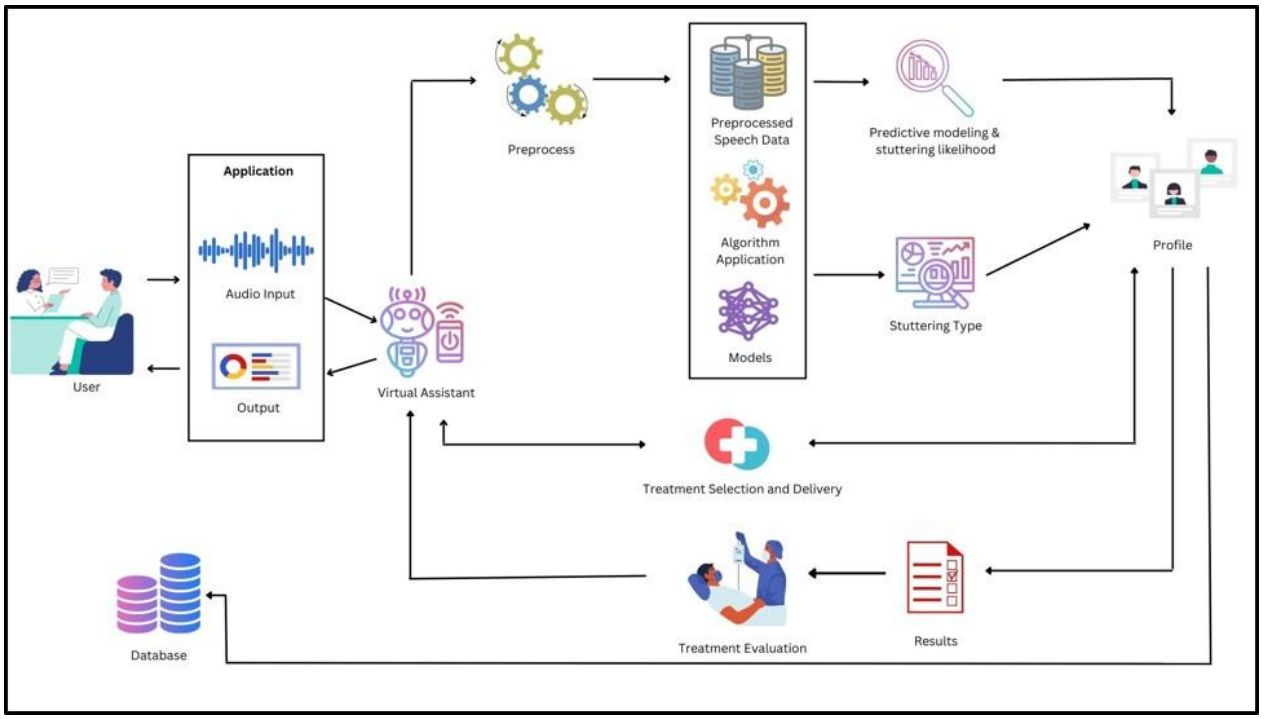


Figure 1: High level system architecture

Data Flow

Raw Audio → [M1 - Elicitation] → 16 kHz PCM → [M2 - Preprocessing] → Clean Segments + Pause Times + SNR Scores → [M3 - Foundation Model Encoder] → 768-dim Embeddings → [M4 - Fluency Extractor] → Confidence-Weighted Features → [M5 - Aggregation] → JSON Feature Vector → [Component 2 + Component 3]

Scalability and Security Considerations

The preprocessing and inference pipeline is containerized using Docker for horizontal scaling. The foundation model encoder is served via ONNX Runtime for reduced inference latency (target < 2 seconds per 60 second audio segment). No raw audio is stored server-side by default; only extracted feature vectors persisted, substantially reducing data sensitivity. All WebSocket connections use TLS 1.3 encryption. Authentication tokens are validated before audio processing begins, consistent with Component 4's privacy preserving federated learning architecture.

5. USER REQUIREMENTS

Stakeholder Identification

- Primary users - Individuals self-administering speech screening (adults with suspected speech disorders).
- Secondary users - Speech language pathologists reviewing screening outputs.
- System administrators - IT staff managing server infrastructure.

Functional Requirements

1. FR1 - The system shall present three speech elicitation tasks in sequence (reading, picture description, spontaneous speech) with clear instructions.
2. FR2 - the system shall capture audio at a minimum of 16 kHz, 16-bit resolution.
3. FR3 - The system shall perform real-time speech quality assessments and display a quality indicator to the user during recording.
4. FR4 - The system shall apply noise suppression preprocessing to all captured audio before feature extraction.
5. FR5 - the system shall extract the following fluency features: WPM, articulation rate, pause frequency, pause duration distribution, and disfluency event density.
6. FR6 - The system shall generate 768-dimensional acoustic embeddings per speech segment using the fine-tuned foundation model.
7. FR7 - The system shall have flag features extracted from audio segments with SNR < 5 dB as low confidence in the output feature vector.
8. FR8 - The system shall transmit the complete feature vector to Component 2 and Component 3 in under 5 seconds of audio completion.
9. FR9 - The system shall function on standard consumer-grade hardware (minimum: dual-core CPU, 4 GB RAM, standard laptop microphone). FR10: The system shall be accessible via modern web browsers without requiring software installation.

Non-Functional Requirements

1. NFR1 (Performance) - End-to-end processing latency from audio submission to feature vector output shall not exceed 5 seconds for a 90-second audio segment.
2. NFR2 (Accuracy) - Fluency feature extraction shall achieve Pearson r > 0.75 against manual annotation on noisy test conditions (15 dB SNR).

3. NFR3 (Usability) - The elicitation interface shall achieve a System Usability Scale (SUS) score ≥ 70 in usability testing with non-technical participants.
4. NFR4 (Security) - All audio data shall be transmitted over TLS 1.3 encrypted WebSocket connections. No raw audio shall be stored server-side beyond the processing session without explicit user consent.

NFR5 (Reliability) - The preprocessing pipeline shall achieve VAD segmentation F1 ≥ 0.92 across all supported noise categories. NFR6 (Compatibility): The web interface shall be compatible with Chrome ≥ 108, Firefox ≥ 110, and Safari ≥ 16.

COMMERCIALIZATION PLAN

Target Market and Customer Persona

1. Primary market - adults with suspected speech disorders in South and Southeast Asia, particularly in Sri Lanka, India, and Bangladesh - regions with the highest SLP-to-population shortfall.
2. Secondary market - SLP clinics and hospitals seek a cost-effective digital triage tool to pre-screen patients before clinical appointments.
3. Tertiary market - educational technology providers developing speech monitoring tools for children with articulation disorders.

Target customer persona - "Priya", a 32-year-old professional in Colombo who has noticed increasing difficulty with fluency over the past year, cannot afford or access an SLP appointment within a reasonable timeframe, and is seeking a trusted, private first-assessment tool accessible from her mobile browser.

Value Proposition

The system provides clinically informed, accessible speech disorder screening to individuals who would otherwise wait 6 - 24 months for a formal SLP appointment. It delivers an actionable screening report - not a diagnosis - within minutes, with privacy-by-design architecture. For clinics, it reduces appointment no-shows and ensures referred patients genuinely require in-depth evaluation.

Revenue Model

Revenue Stream	Pricing	Target Segment
Freemium SaaS	Free basic - LKR 500 per month premium	Individual users in Sri Lanka and South Asia seeking early screening
B2B Institutional License	LKR 150,000 - 300,000 per year	Hospitals and SLP clinics integrating screening into patient intake workflows
Developer API Access	USD 0.05 -0.10 per screening query	EdTech platforms embedding speech assessment in learning applications

Table 5: Revenue Model

Cost Estimation

Cost Item	Estimated Monthly (LKR)
Cloud compute (GPU inference, AWS g4dn.xlarge)	3,000
Storage (S3, feature vectors only)	2,000
CDN and bandwidth	1,000
Domain and SSL	3500
Total	9500/months

Table 6: Cost Estimation

Competitive Advantage

No existing commercial speech screening tool incorporates ASR uncertainty aware fluency measurement or performs multi-task elicitation within a single browser session. The system's noise robustness benchmarking also provides a documented performance guarantee under realistic conditions - a claim no competitor currently makes.

Intellectual Property Considerations

The ASR-uncertainty-aware fluency metric weighting methodology and the adaptive speech quality gating mechanism are novel algorithmic contributions suitable for provisional patent filing. The fine-tuned foundation model weights may be released under an open research license (CC BY-NC 4.0) to build academic credibility while protecting commercial deployment rights.

7. BUDGET AND JUSTIFICATION

Cost Item	Estimated Cost	Justification
Cloud GPU compute (AWS g4dn.xlarge, 3 months fine-tuning)	**LKR 4000**	HuBERT-Large fine-tuning requires GPU
Cloud inference hosting (6 months development + testing)	**LKR 10,000**	Real-time ONNX inference serving during development and evaluation
Cloud storage (model checkpoints, feature vectors)	**LKR 700**	Checkpoints 3 GB - feature vector archive 10 GB
Software licenses	**LKR 0**	All tools open source, PyTorch, HuggingFace, Librosa, React.js
Corpus access (TORGO, SEP-28k, UASPEECH)	**LKR 0**	Publicly available for academic research, no licensing cost
Testing hardware (condenser + dynamic microphones)	**LKR 4500**	Required for cross-microphone robustness evaluation experiments
Contingency (10%)	**LKR 13,000**	Unexpected computer overruns or additional storage requirements
TOTAL	**LKR 32,000**	

Table 7: Budget and Justification

8. WORK BREAKDOWN STRUCTURE (WBS)

WBS ID	Task Description	Sub-task Detail	Timeline
WBS 1.1	**Requirements & Design**		**Weeks 1–3**
1.1.1		Review and finalize speech elicitation task selection (passage, picture stimulus, spontaneous prompt)	Weeks 1–2
1.1.2		Define feature vector schema and API contract with Component 2 and Component 3	Weeks 2–3
1.1.3		Design preprocessing pipeline architecture and module interfaces	Weeks 2–3
1.1.4		Complete literature review and research gap documentation	Weeks 1–3
WBS 1.2	**Data Preparation & Model Setup**		**Weeks 4–7**
1.2.1		Download, preprocess, and verify TORGO, SEP-28k, UASPEECH corpora	Weeks 4–5
1.2.2		Implement noise data augmentation pipeline with MUSAN at multiple SNR levels	Weeks 5–6
1.2.3		Set up HuBERT and Whisper fine-tuning environments (GPU, HuggingFace Trainer)	Weeks 5–6
1.2.4		Conduct baseline evaluation of pre-trained models (no fine-tuning) as performance reference	Weeks 6–7
WBS 1.3	**Preprocessing Pipeline**		**Weeks 5–9**
1.3.1		Integrate Silero-VAD; validate against manual segmentation ground truth	Weeks 5–6

1.3.2		Implement Wiener filter spectral noise suppression module	Weeks 6–7
1.3.3		Implement WADA-SNR speech quality assessment gate and flagging logic	Weeks 8–9
1.3.4		Write and run unit tests for each preprocessing stage in isolation	Week 9
WBS 1.4	**Foundation Model Fine-Tuning**		**Weeks 8–13**
1.4.1		Fine-tune HuBERT-Large on TORGO + UASPEECH (dysarthria classification objective)	Weeks 8–11
1.4.2		Fine-tune Whisper-Medium on SEP-28k (fluency/disfluency detection objective)	Weeks 10–12
1.4.3		Hyperparameter optimization and model selection on validation split	Weeks 12–13
1.4.4		Evaluate embedding quality via downstream disorder classification accuracy	Week 13
WBS 1.5	**Fluency Feature Extraction**		**Weeks 11–16**
1.5.1		Implement Whisper token-level log-probability confidence score extraction	Weeks 11–12
1.5.2		Implement ASR confidence-weighted WPM and articulation rate computation	Weeks 13–14
1.5.3		Implement transcription-independent pause and silence features from VAD timestamps	Weeks 14–15
1.5.4		Validate all fluency features against manual annotation; compute inter-rater reliability	Weeks 15–16

WBS 1.6	**Elicitation Interface**		**Weeks 14–18**
1.6.1		Implement React.js task administration UI with timing, instructions, and waveform visualizer	Weeks 14–16
1.6.2		Integrate WebAudio API microphone capture and WebSocket audio streaming	Weeks 16–17
1.6.3		Implement real-time SNR quality indicator (green/amber/red) in UI	Week 17
1.6.4		Conduct usability testing (n ≈ 15 participants) and compute SUS score	Weeks 17–18
WBS 1.7	**Robustness Evaluation & Integration**		**Weeks 17–22**
1.7.1		Conduct cross-corpus noise robustness experiments (6 SNR levels × 3 noise types)	Weeks 17–19
1.7.2		Analyze feature stability and document robustness benchmarks	Weeks 19–20
1.7.3		Integration testing with Component 2 (Alignment and Articulation Analytics)	Weeks 20–21
1.7.4		Integration testing with Component 3 (Multi-Disorder Severity Classification)	Weeks 21–22
WBS 1.8	**Documentation & Reporting**		**Weeks 20–24**
1.8.1		Write technical documentation (API reference, pipeline docs, model cards)	Weeks 20–22
1.8.2		Prepare and submit interim progress report	Weeks 22–23

1.8.3		Prepare and submit final component evaluation report	Weeks 23–24

Table 8: Work breakdown structure

9. GANTT CHART

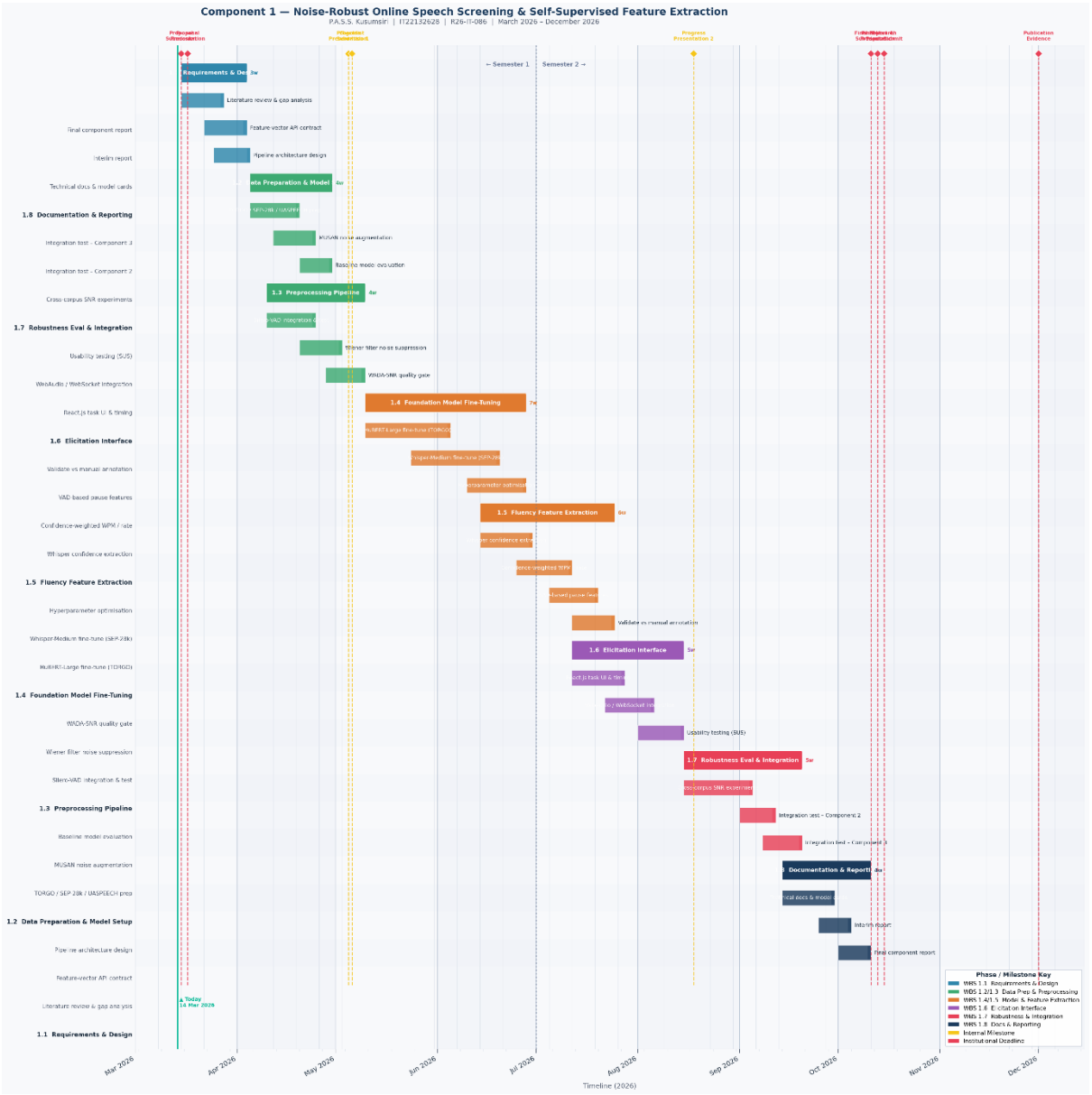


Figure 2: Grantt chart

10.REFERENCES LIST

[1] World Health Organization, "Deafness and hearing loss," WHO Fact Sheet, Apr. 2021. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/deafness-and-hearing-loss

[2] N. J. Bhana, "Global prevalence of communication disorders," Int. J. Lang. Commun. Disord., vol. 56, no. 3, pp. 512–524, 2021.

[3] World Health Organization, "Rehabilitation in health systems," WHO, Geneva, Switzerland, Tech. Rep., 2017.

[4] E. Yairi and N. Ambrose, "Epidemiology of stuttering: 21st century advances," J. Fluency Disord., vol. 38, no. 2, pp. 66–87, 2013.

[5] L. J. Yorkston, R. M. Miller, and E. A. Strand, Management of Speech and Swallowing in Degenerative Diseases, 3rd ed. Austin, TX: Pro-Ed, 2004.

[6] American Institute for Stuttering, "Economic impact of stuttering," AIS Report, 2020.

[7] F. Rudzicz, "Adjusting dysarthric speech signals to be more intelligible," Comput. Speech Lang., vol. 28, no. 6, pp. 1163–1177, 2014.

[8] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2020, pp. 12449–12460.

[9] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhotia, R. Salakhutdinov, and A. Mohamed, "HuBERT: Self-supervised speech representation learning by masked prediction of hidden units," IEEE/ACM Trans. Audio, Speech, Lang. Process., vol. 29, pp. 3451–3460, 2021.

[10] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in Proc. Int. Conf. Mach. Learn. (ICML), 2023, pp. 28492–28518.

[11] F. Rudzicz, A. K. Namasivayam, and T. Wolff, "The TORGO database of acoustic and articulatory speech from speakers with dysarthria," Lang. Resources Eval., vol. 46, no. 2, pp. 523–541, 2012.

[12] K. Lea, M. Al-Ghezi, R. Barber, A. Schwartz, J. Bergelson, and Z. Frisby, "SEP-28k: A dataset for stuttering event detection from podcasts with people who stutter," in Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP), 2021, pp. 6798–6802.

[13] H. Kim, M. Hasegawa-Johnson, A. Perlman, J. Gunderson, T. S. Huang, K. Watkin, and S. Frame, "Dysarthric speech database for universal access research," in Proc. Interspeech, 2008, pp. 1741–1744.

[14] S. P. Bayerl, K. A. Riedhammer, and T. Bocklet, "Detecting dysfluencies in stuttering therapy using wav2vec 2.0," in Proc. Interspeech, 2022, pp. 1963–1967.

[15] N. Vaessen and D. A. van Leeuwen, "Fine-tuning wav2vec2 for stuttering detection," arXiv preprint arXiv:2107.13484, 2021.

[16] A. Illa and P. K. Ghosh, "Low resource automatic severity assessment of dysarthric speech using transfer learning," in Proc. Interspeech, 2020, pp. 4401–4405.

[17] C. Botelho, T. Schultz, and P. Teixeira, "Pathological speech detection using x-vector embeddings," in Proc. Interspeech, 2022, pp. 2213–2217.

[18] J. R. Orozco-Arroyave et al., "Automatic detection of Parkinson's disease in running speech: Exploiting complementary features," J. Acoust. Soc. Am., vol. 136, no. 5, pp. 2644–2654, 2014.

[19] I. Lian, B. Tian, and L. Shi, "Unconstrained stuttering detection and severity estimation with transformer encoders," in Proc. Interspeech, 2023, pp. 856–860.

[20] F. L. Darley, A. E. Aronson, and J. R. Brown, Motor Speech Disorders. Philadelphia, PA: W.B. Saunders, 1975.

[21] I. Silero Team, "Silero VAD: Pre-trained enterprise-grade voice activity detector," GitHub, 2021. [Online]. Available: https://github.com/snakers4/silero-vad

[22] C. H. Taal, R. C. Hendriks, R. Heusdens, and J. Jensen, "An algorithm for intelligibility prediction of time-frequency weighted noisy speech," IEEE Trans. Audio, Speech, Lang. Process., vol. 19, no. 7, pp. 2125–2136, 2011

11.APPENDICES

Appendix A - Risk Analysis

Risk	Likelihood	Impact	Mitigation Strategy
Dataset licensing restrictions (TORGO, SEP-28k)	**Low**	**High**	Datasets confirmed for academic use, cite original papers and follow redistribution terms
GPU compute budget overrun during fine-tuning	**Medium**	**Medium**	Use mixed-precision (FP16) training; limit fine-tuning to 15 epochs with early stopping
ASR confidence scores unavailable in target Whisper version	**Low**	**High**	Use Whisper v3 which exposes log-prob natively; fall back to CTC confidence from wav2vec 2.0
Insufficient usability testing participants recruited	**Medium**	**Low**	Recruit within SLIIT IT faculty; supplement with think-aloud sessions ($n \geq 8$ sufficient for SUS)
Integration of API mismatch with Component 2 or Component 3	**Medium**	**Medium**	Define and freeze JSON feature vector API schema by Week 4; use contract testing throughout
Noisy audio quality from participant microphones in usability test	**High**	**Low**	SNR quality gate flags low-quality segments, usability test evaluates UX, not clinical accuracy

Table 9: Appendix A - Risk Analysis

Appendix B - Preliminary UI Mockup Notes

The elicitation interface should present tasks sequentially with a visual countdown timer. A waveform visualizer provides real-time audio feedback. A color-coded SNR indicator (green/amber/red) informs users of recording quality before submission. Task instructions are displayed in plain language with audio read-aloud option for accessibility.

Appendix C - Ethical Clearance

Application to SLIIT Ethics Review Committee to be submitted before participant usability testing. No clinical data from disorder populations will be collected directly; all disorder-speech training data derives from pre-approved public corpora.

Component: Alignment & Articulation Analytics

Project Proposal Report

Student Name – IT22257086

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Randiw E.Y	**IT 22257086**	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Ms.Chathurangika Kahandawaarachchi		

ABSTRACT (Maximum 300 words)

Speech disorders such as stuttering, dysarthria, and articulation impairments affect millions of individuals worldwide, yet access to timely and consistent speech assessment remains limited, particularly in regions with scarce clinical resources. While recent advances in artificial intelligence have enabled automated speech analysis and disorder detection, most existing systems focus primarily on fluency features or high level acoustic patterns, often overlooking the detailed phonetic articulation analysis required for accurate clinical interpretation. This proposal presents the **Alignment and Articulation Analytics Component**, a core subsystem of the **AI Powered Speech Disorder Screening and Therapy Platform**, designed to provide fine grained phoneme level analysis of English speech for improved disorder assessment and therapy guidance.

The component addresses several limitations in current automated speech analysis systems, the absence of reliable phoneme level articulation analysis in real world environments, the limited integration of forced alignment techniques within AI driven clinical support tools, the lack of interpretable articulation metrics for clinicians and users, and the absence of uncertainty aware alignment evaluation mechanisms. By implementing a phoneme level forced alignment pipeline capable of measuring phoneme duration, articulation timing variation, and pronunciation accuracy, the component enables detailed detection of articulation inconsistencies and mispronunciations associated with speech disorders. In addition, the system incorporates alignment of confidence scoring to estimate boundary uncertainty and improve the reliability of articulation assessment results.

The research adopts data **driven speech processing and machine learning methodology**, combining phonetic alignment techniques with statistical articulation analytics. The implementation utilizes technologies such as **Python, PyTorch, and Montreal Forced Aligner**, alongside signal processing libraries for speech feature extraction and analysis. Expected outcomes include an interpretable articulation scoring framework capable of supporting automated speech disorder screening and contributing phonetic insights to the platform's multi disorder severity classification engine.

This research contributes to **SDG 3 (Good Health and Well-being)** by improving access to speech health screening technologies and supports **SDG 10 (Reduced Inequalities)** by enabling scalable and accessible AI-driven speech assessment solutions for underserved communities.

Keywords: Speech Disorder Analysis, Phoneme Alignment, Articulation Analytics, Forced Alignment, Speech Processing, AI-Assisted Speech Therapy

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6. UA-Speech (https://www.kaggle.com/datasets/aryashah2k/noise-reduced-uaspeech-dysarthria-dataset) for cross-dataset validation

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ASR	Automatic Speech Recognition
CTC	Connectionist Temporal Classification
DAF	Delayed Auditory Feedback
DL	Deep Learning
DSR	Design Science Research
GPU	Graphics Processing Unit
HuBERT	Hidden Unit Bidirectional Encoder Representations from Transformers
ML	Machine Learning
MUSAN	Music, Speech, and Noise corpus
NFR	Non-Functional Requirement
ONNX	Open Neural Network Exchange
PCM	Pulse Code Modulation
SDG	Sustainable Development Goal
SLP	Speech-Language Pathologist / Pathology
SNR	Signal-to-Noise Ratio
SUS	System Usability Scale
TLS	Transport Layer Security

VAD	Voice Activity Detection
WADA	Waveform Amplitude Distribution Analysis
WPM	Words Per Minute

Table 10:List of Abbreviations

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7. SEP-28k (https://www.kaggle.com/datasets/ikrbasak/sep-28k) for stuttering event detection
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9. UA-Speech (https://www.kaggle.com/datasets/aryashah2k/noise-reduced-uaspeech-dysarthria-dataset) for cross-dataset validation

1. INTRODUCTION

1.1 Background and Context

Speech disorders such as stuttering, dysarthria, and articulation impairments affect millions of people worldwide, yet timely assessment and therapy remain limited, particularly in regions with scarce clinical resources. Traditional speech evaluations rely heavily on in person expert analysis, which is often time consuming, subjective, and difficult to scale.

Recent advances in artificial intelligence have opened new opportunities for automated speech analysis. Many existing systems focus on high level acoustic patterns or fluency features, providing useful but coarse insights. However, these approaches often overlook the detailed phoneme level articulation analysis that clinicians rely on to diagnose and plan therapy effectively. Without fine grained analysis, subtle mispronunciations, timing inconsistencies, and other articulation irregularities may go undetected.

The **Alignment and Articulation Analytics Component** address these gaps by providing precise, phoneme level analysis of English speech. Using a forced alignment pipeline, the system measures phoneme durations, timing variations, and pronunciation accuracy. It also incorporates alignment confidence scoring to quantify uncertainty, ensuring clinicians and users can trust the automated assessments.

This component integrates seamlessly into a broader AI powered speech disorder assessment platform, offering interpretable metrics that support early detection, personalized therapy guidance, and longitudinal tracking of progress. By combining speech processing techniques, machine learning, and statistical articulation analytics, the component brings clinical level insight into an accessible, scalable system.

SDG Alignment: This research supports **SDG 3 (Good Health and Well-being)** by improving access to speech health screening technologies, and **SDG 10 (Reduced Inequalities)** by enabling AI driven assessment for underserved communities.

1.2 Literature Review

1.2.1 *Speech Disorder Detection Using AI*

Artificial intelligence has increasingly been applied to the detection and diagnosis of speech disorders. Modern AI systems use acoustic features, prosody, and fluency patterns to classify speech abnormalities. These systems can assist clinicians by providing faster, more consistent assessments than manual evaluation alone. However, most AI approaches focus on high level speech patterns rather than fine grained phonetic details, limiting their usefulness in identifying subtle articulation errors.

1.2.2 *Phoneme Level Speech and Forced Alignment*

Phoneme level analysis examines speech at the smallest meaningful sound units, enabling detailed insights into pronunciation accuracy, timing, and consistency. Forced alignment techniques, such as the Montreal Forced Aligner (MFA), automatically map audio to phonetic transcripts, providing accurate boundaries for each phoneme. While widely used in research and linguistics, forced alignment has been underutilized in AI driven clinical applications, where alignment accuracy and confidence are critical for reliable assessments.

1.2.3 *Articulation Assessment in Speech Therapy Systems*

Articulation assessment is a core component of speech therapy. Traditional systems rely on expert observation to identify mispronunciations and timing inconsistencies. Some automated therapy tools provide feedback using visual or audio cues, but few integrate interpretable, quantitative metrics for each phoneme. Incorporating such metrics allows clinicians to track progress objectively and enables AI systems to support personalized therapy planning.

1.2.4 *Machine Learning in Clinical Speech Analysis*

Machine learning techniques, including deep learning and statistical modeling, have been applied to predict speech disorder severity, classify speech types, and generate therapy recommendations. These approaches require large datasets and robust feature extraction methods. Combining ML with phoneme level alignment and articulation analysis can improve accuracy and interpretability, enabling systems that are both clinically meaningful and scalable.

1.3 Research Gap

Despite the advances described above, several gaps remain:

- Existing AI systems rarely provide **phoneme-level articulation analysis** in real-world environments.
- Forced alignment techniques are **under-integrated** into clinical support tools, limiting reliability and interpretability.
- There is a **lack of standardized metrics** for measuring misarticulations and pronunciation accuracy.
- Systems rarely account for **alignment uncertainty**, which can reduce confidence in automated assessments.

2. OBJECTIVE

2.1 Main Objective

To design, implement, and evaluate the Alignment and Articulation Analytics Component that provides phoneme level analysis of English speech, measures articulation accuracy, phoneme timing, and mispronunciation patterns, incorporates alignment confidence scoring to quantify uncertainty, generates interpretable metrics for clinicians and users, and supports automated speech disorder assessment and therapy guidance within a scalable AI powered platform.

2.2 Specific Objectives

- SO**1:** To design and implement a phoneme level forced alignment model that maps speech audio to text with accurate phoneme boundaries, capturing detailed articulation patterns across speech samples.
- SO**2:** To extract articulation features, including phoneme duration, timing variation, and pronunciation accuracy, enabling quantitative analysis of speech production.
- SO**3:** To develop a misarticulation detection and phonetic scoring mechanism that identifies pronunciation errors and inconsistencies associated with speech disorders.
- SO**4:** To implement alignment confidence and boundary uncertainty analysis, providing interpretable reliability metrics for automated assessments.
- SO**5:** To integrate the component with the multi disorder severity classification system, supporting longitudinal monitoring and AI assisted therapy recommendations.

3. METHODOLOGY

3.1 Research Approach

This component adopts data **driven speech processing and analytical research methodology** supported by expert evaluation. The primary objective is to analyze speech signals at the phoneme level to identify articulation patterns and mispronunciations associated with speech disorders.

Unlike purely predictive AI models that focus only on classification accuracy, this component emphasizes **interpretable phonetic analysis**. The system performs forced alignment between speech audio and phonetic transcripts to measure articulation timing, pronunciation accuracy, and phoneme duration variations.

Quantitative validation will evaluate the reliability of phoneme alignment and articulation metrics. Statistical analysis will be used to measure pronunciation deviations and timing inconsistencies across speech samples. Performance evaluation will include alignment accuracy, articulation scoring consistency, and misarticulation detection accuracy.

In addition, qualitative evaluation will be conducted through **expert review by speech specialists**, assessing whether the generated articulation insights and pronunciation feedback are clinically meaningful and useful for speech therapy support.

3.2 Technical Methods and Algorithms

3.2.1 *Phoneme Level Forced Alignment Model*

The system will implement a phoneme-level forced alignment pipeline using the **Montreal Forced Aligner (MFA)**. The alignment model synchronizes recorded speech audio with phonetic transcripts to determine precise start and end timestamps for each phoneme.

This alignment process enables the system to generate detailed phonetic segmentation of speech signals, allowing accurate measurement of articulation patterns and timing characteristics. The resulting phoneme boundaries form the foundation for further articulation analysis.

3.2.2 *Articulation Feature Extraction and Timing Analysis*

Once phoneme alignment is obtained, the system extracts articulation related speech features including:

- *phoneme duration*
- *articulation timing variation*
- *pronunciation accuracy*
- *speech rate and rhythm characteristics*

Signal processing techniques will be applied to extract acoustic features from speech signals using libraries such as Librosa and Praat-based processing tools. These features provide quantitative indicators of articulation quality and help identify deviations from expected pronunciation patterns.

3.2.3 *Misarticulation Detection and Phonetic Scoring Mechanisms*

A phonetic scoring mechanism will be implemented to evaluate pronunciation accuracy by comparing expected phoneme patterns with the detected phoneme outputs from the alignment process.

The system will identify potential misarticulations by analyzing deviations in phoneme timing, pronunciation consistency, and acoustic similarity. A scoring model will generate interpretable articulation scores for each phoneme and for entire speech samples.

These scores provide clinicians and users with structured insights into pronunciation strengths and weaknesses, supporting targeted speech therapy guidance.

3.2.4 *Alignment Confidence and Boundary Uncertainty Analysis*

To improve reliability, the system will incorporate **alignment confidence scoring** that estimates uncertainty in phoneme boundary detection.

Confidence scores will be generated by analyzing acoustic likelihoods and alignment stability across phoneme segments. Low confidence alignments will be flagged for further review, ensuring that articulation analysis results remain trustworthy and interpretable.

This uncertainty aware approach increases the reliability of automated speech assessment systems used in real-world environments.

3.3 Tools and Platforms

Tool / Platform	Purpose in the Research
Python	Primary programming language used for implementing speech processing algorithms and machine learning
PyTorch	Deep learning framework used to build and train speech analysis models
Montreal Forced Aligner	Tool used to perform phoneme level forced alignment between speech audio and transcripts
Librosa	Audio processing library used for extracting speech features such as MFCC (Mel Frequency Cepstral Coefficients) and spectral characteristics
Praat	Speech analysis software used to analyze phonetic features and validate articulation measurements
React.js	Frontend framework used to develop the user interface for speech analysis visualization
Node.js	Backend environment used to manage API services and system integration
MongoDB	Database used to store speech recordings, extracted features, and analysis results
GitHub	Version control and collaboration platform used for project development and code management

Table 11:Tools and Platforms

3.4 Data Collection and Ethical Considerations

During the development phase, the system will use **publicly available speech datasets** and simulated speech samples to test alignment and articulation analysis algorithms.

For any real user data collection:

1. Ethical approval will be obtained prior to data collection.
2. Participants will provide **informed consent** before submitting speech recordings.
3. Personal identifiers will be removed from speech datasets.
4. Speech recordings will be securely stored with encrypted identifiers.
5. Data usage will follow established research ethics and privacy guidelines.

3.5 Validation Strategy and Performance Metrics

The effectiveness of the component will be evaluated using several performance indicators.

- **Alignment Accuracy**
  Accuracy of phoneme boundary detection compared with reference alignments from validated speech datasets.
- **Articulation Scoring Reliability**
  Consistency of articulation scores across repeated speech samples from the same speaker.
- **Misarticulation Detection Accuracy**
  Accuracy of identifying pronunciation errors compared with expert annotated speech samples.
- **Expert Evaluation**
  Speech experts will review generated articulation reports and rate the usefulness of the system's feedback.
- **System Performance**
  Speech analysis processing time will be measured to ensure efficient system performance during real time or near real time speech assessment

4. HIGH LEVEL SYSTEM ARCHITECTURE

4.1 Component Overview

The **Alignment and Articulation Analytics Component** functions as the phonetic analysis layer of the AI Powered Speech Disorder Screening and Therapy Platform. Its primary role is to analyze recorded speech at the phoneme level and generate detailed articulation insights that can support automated disorder screening and therapy guidance.

The component receives speech recordings and phonetic transcripts from the platform's input modules and processes them through a forced alignment pipeline. It then extracts articulation features, evaluates pronunciation accuracy, and produces interpretable articulation metrics that can be used by clinicians, researchers, and the system's disorder classification engine.

4.2 Speech Processing Pipeline

The speech processing pipeline is responsible for transforming raw speech recordings into detailed phoneme level articulation insights. The pipeline begins with speech input collected through the platform interface, where users submit audio recordings of spoken text.

The first stage of the pipeline performs **audio preprocessing**, which includes noise reduction, normalization, and segmentation of the speech signal. These steps ensure that the audio data is clean and suitable for further phonetic analysis.

Next, the processed audio is passed to the **phoneme level forced alignment module**. This module uses the Montreal Forced Aligner to synchronize the speech signal with its corresponding phonetic transcription. The alignment process determines the start and end boundaries of each phoneme within the speech recording.

Following alignment, the system performs **articulation feature extraction**. In this stage, the system calculates phoneme duration, articulation timing variation, pronunciation accuracy, and other relevant acoustic characteristics. These features provide quantitative indicators of how accurately each phoneme is produced.

Finally, the processed articulation data is used by the **misarticulation detection mechanism**, which identifies pronunciation inconsistencies and generates articulation scores. The results are stored in the system database and forwarded to other platform components for further analysis.

4.3 Integration with Multi disorder Severity Classification Module

The articulation analytics component integrates with the platform's **multi disorder severity classification module**, which is responsible for assessing the overall severity of speech disorders.

The phoneme level articulation metrics generated by this component serve as input features for the classification system. These metrics include pronunciation accuracy scores, phoneme duration statistics, articulation consistency indicators, and detected misarticulation patterns.

By incorporating these detailed phonetic insights, the severity classification module can make more informed predictions regarding the presence and intensity of speech disorders such as stuttering, dysarthria, or articulation impairments.

This integration enables the platform to combine **phonetic level articulation analysis with machine learning based disorder classification**, providing a comprehensive assessment of speech performance.

4.4 Scalability and Security

The system architecture is designed to support scalable speech processing while maintaining strong data security.

To support multiple users and speech recordings, the platform will use a structured database that efficiently stores speech analysis results and articulation metrics. The architecture allows multiple speech samples to be processed simultaneously without performance degradation.

From a security perspective, all speech recordings and analysis outputs will be stored using secure data management practices. Sensitive information will be protected through encryption and controlled access mechanisms. Only authorized users will have access to speech recordings and articulation reports.

Additionally, when speech data is used for research or analysis, personal identifiers will be removed and replaced with anonymized identifiers. These measures ensure that user privacy is protected while enabling reliable speech disorder analysis.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Several stakeholders interact with the Alignment and Articulation Analytics component.

• **Primary stakeholders:** Speech therapists and clinicians who use articulation analysis results to evaluate pronunciation patterns and support therapy decisions.

• **Secondary stakeholders:** Individuals undergoing speech assessment who receive feedback on their speech articulation and pronunciation accuracy.

• **Research stakeholders:** Speech researchers and developers interested in analyzing phonetic patterns and improving speech disorder detection systems.

• **Technical stakeholders:** Other components of the AI speech disorder platform that consume articulation metrics to perform disorder classification and therapy recommendation generation.

5.2 Functional Requirements

ID	Functional Requirement Description
1	The system records speech audio submitted by users through the platform interface.
2	The system performs phoneme level forced alignment between speech audio and phonetic transcripts.
3	The system extracts articulation features such as phoneme duration, pronunciation accuracy, and articulation timing variation.
4	The system detects misarticulations and pronunciation inconsistencies in speech recordings.
5	The system calculates articulation scores for individual phonemes and entire speech samples.
6	The system generates articulation analysis reports for clinicians and users.
7	The system stores speech analysis results for further processing and integration with the severity classification module.
8	The system provides visualization of phoneme segmentation and articulation metrics through the user interface.

Table 12:Functional Requirements

5.3 Non-Functional Requirements

The system must satisfy several quality and performance requirements.

- **Performance**
  Speech analysis should be completed within a few seconds for standard speech recordings.
- **Accuracy**
  Phoneme alignment and articulation feature extraction should provide reliable and consistent results when tested using benchmark speech datasets.
- **Usability**
  The system interface should present articulation analysis results in a clear and understandable format for clinicians and non-technical users.
- **Security and Privacy**
  Speech recordings and user data must be securely stored with encryption and controlled access.
- **Scalability**
  The system should support multiple users submitting speech recordings simultaneously without performance issues.
- **Reliability**
  Speech processing modules should produce consistent analysis results across repeated speech inputs.

6. COMMERCIALIZATION PLAN

6.1 Target Market

The primary market for this system includes speech therapy clinics, healthcare institutions, and educational organizations that provide speech disorder assessment and therapy services.

The platform may also benefit telehealth providers and mobile health applications that aim to deliver remote speech therapy support. These solutions are particularly valuable in regions where access to speech therapy specialists is limited

6.2 Value Proposition

The Alignment and Articulation Analytics component offer a scalable solution for automated speech disorder assessment. By analyzing speech at the phoneme level, the system provides detailed articulation insights that traditional speech analysis tools often lack.

The integration of forced alignment, articulation feature extraction, and AI based disorder classification allows clinicians to obtain objective pronunciation analysis and monitor speech therapy progress more effectively.

This approach improves accessibility to speech assessment technologies while supporting more accurate and interpretable speech disorder evaluation.

6.3 Revenue Model

The platform can generate revenue through several models.

• **Software subscription:** Healthcare institutions and speech therapy clinics can subscribe to the platform for continuous speech assessment services.

• **Integration services:** The articulation analytics module can be integrated into telehealth platforms and digital therapy applications through API access.

• **Educational licensing:** Universities and training institutes may use the platform for speech pathology research and teaching purposes.

• **Research partnerships:** Collaboration with research institutions and healthcare organizations for speech disorder studies.

6.4 Intellectual Property Considerations

- The system includes several innovative components that may have intellectual property value.
- These include the phoneme-level articulation analytics pipeline, the integration of forced alignment with articulation scoring, and the visualization methods used for interpreting speech articulation patterns.
- Protecting these innovations through appropriate intellectual property strategies can support future commercialization and technology development.

7. BUDGET AND JUSTIFICATION

Item	Cost (LKR)	Justification
Cloud GPU compute (AWS, 3 months fine-tuning)	4000	HuBERT-Large fine-tuning requires GPU, estimated 120 hours at
Cloud inference hosting (6 months development + testing)	10 000	g4dn.xlarge for real-time ONNX inference serving
Storage (model checkpoints, feature vectors)	700	Model checkpoints ~3 GB; feature vector archive ~10 GB
Software licenses	0	All tools open source (PyTorch, HuggingFace, Librosa, React)
Corpus access (TORGO, SEP-28k, UASPEECH)	0	Publicly available for academic research
Microphone for testing (condenser + dynamic)	4500	Required for cross-microphone robustness evaluation
Contingency (10%)	13 000	Unexpected compute overruns
Total	**32 000**	

Table 13:Budget and Justification

8. WORK BREAKDOWN STRUCTURE (WBS)

Phase	Task Description
1	Literature review and requirement analysis
2	System architecture design and speech processing pipeline design
3	Implementation of phoneme-level forced alignment module
4	Development of articulation feature extraction algorithms
5	Implementation of misarticulation detection and phonetic scoring
6	Integration with multi-disorder severity classification module
7	Testing and validation using speech datasets
8	System evaluation and performance analysis
9	Documentation and final project report preparation

Table 14:Work Breakdown Structure

9. GANTT CHART

Project spans two semesters (approximately 28 weeks):

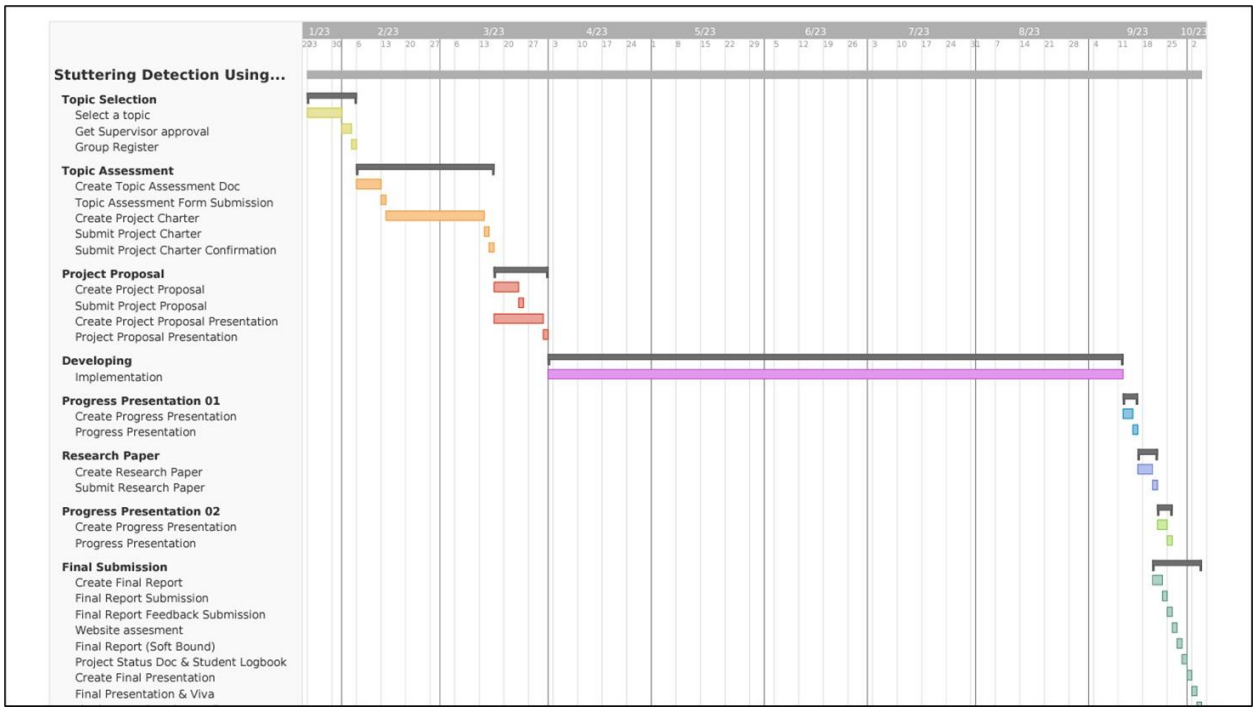


Figure 3:Grantt Chart

10. REFERENCES LIST

[1] M. McAuliffe, M. Socolof, S. Mihuc, M. Wagner, and M. Sonderegger, *Montreal Forced Aligner: Trainable Text-Speech Alignment Using Kaldi*, Interspeech 2017, pp. 498–502.

[2] X. Zhang, J. Wu, and C. Lee, "Automatic Speech Recognition for Clinical Speech Assessment," *Computer Speech & Language*, vol. 66, 2021, pp. 101–120.

[3] Y. Kim, J. H. Park, and H. Lee, "Phoneme-level articulation analysis using deep neural networks for speech disorder detection," *IEEE Access*, vol. 9, pp. 145678–145690, 2021.

[4] D. Jurafsky and J. H. Martin, *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition*, 3rd ed., Pearson, 2021.

[5] L. Goldstein, J. Byrd, and M. Saltzman, "The role of articulatory phonology in speech disorder analysis," *Clinical Linguistics & Phonetics*, vol. 19, no. 6-7, pp. 545–567, 2005.

[6] R. Villing, S. Ward, and T. Scott, "Forced alignment for dysarthric speech: Challenges and solutions," *Speech Communication*, vol. 100, pp. 1–14, 2018.

[7] S. Davis and P. Mermelstein, "Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 28, no. 4, pp. 357–366, 1980.

[8] S. Barocas, M. Hardt, and A. Narayanan, *Fairness and Machine Learning*, Cambridge, MA: fairmlbook.org, 2019.

[9] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778. *(Used as reference for deep learning frameworks applied in speech analysis.)*

[10] UN SDG, "Sustainable Development Goals," United Nations, 2015. [Online]. Available: https://sdgs.un.org/goals

11. APPENDICES

Appendix A: Risk Analysis

Risk ID	Risk Description	Likelihood	Impact	Mitigation Strategy
R1	Poor audio quality from user recordings leading to inaccurate phoneme alignment	Medium	High	Implement pre-processing filters, denoising algorithms, and user guidelines for recording quality
R2	Forced alignment errors due to non-standard accents or speech impairments	Medium	High	Use multiple alignment models, confidence scoring, and human expert verification for low-confidence alignments
R3	Insufficient or imbalanced dataset for training/testing	Medium	High	Augment data using synthetic samples, publicly availlable datasets, and cross-validation techniques
R4	Privacy breach of user speech data	Low	High	Encrypt all data at rest (AES-256), anonymize identifiers, implement strict access controls
R5	System performance degradation with large numbers of concurent users	Low	Medium	Optimize database queries, partition profiles by user ID, implement caching, and stress-test system
R6	Misinterpretation of statistical or articulation scores by users	Medium	Medium	
R7	Ethical or fairness concerns (bias in alignment or scoring)	Low	High	Include clear visualizations, confidence intervals, and expert-reviewed explanations in UI
R8	Integration failures with multi-disorder severity classification module	Low	Medium	Develop API tests, error logging, and modular component design for easier debugging

Appendix B: Speech Dataset and Feature Extraction Summary

Dataset / Source	Description	Features Extracted	Purpose
Public Speech Dataset	Publicly available speech recordings containing clear phoneme	Phoneme boundaries, phoneme duration, pronunciation accuracy	Training and validating the forced aligmnent model
Speech Disorder Dataset	Speech recordings from individuals with articulation or pronunciation difficulties.	Misarticulation patterns, phoneme substitution errors, timing irregularities	Evaluating misarticulatin detection capability
Synthetic Speech Samples	Generated speech data used for controlled testing scenarios	Phoneme segmentation accuracy, articulation timing consistency	Testing robustness of the aligmment algorithm
User Recorded Speech	System performance degradation with large, numbers of concurrent users	Phoneme timing variation, articulation score, pronunciation colnfio	Optimize dalabase queries, partfion profilee by, user iD, implement and sffess-test-system

AI Online Speech Disorder Screening and Therapy Assistant

R26-IT-086

Component:

Multi-Disorder Severity Classification and Risk Prediction

Project Proposal Report

I.L.S. Tharushan

IT21315114

B.Sc. (Hons) Degree in Information Technology

Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
I.L.S. Tharushan	**IT21315114**	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
MS. Chathurangika Kahandawaarachchi	14 / 03 / 2026	

ABSTRACT

With 70 million patients worldwide living with speech disorders like stuttering, dysarthria, and articulation disorders, the clinical and social consequences are considerable. In developing countries like Sri Lanka, the scarcity of medical facilities makes the situation worse. It is imperative to assess the severity of the disorder accurately to provide timely therapy. However, the clinical community is currently unable to do so due to the lack of speech-language pathologists and geographical barriers. Moreover, the tools currently available are only useful for binary classification of a single disorder under a controlled acoustic environment. Therefore, the automated tools currently available are not useful for real-time clinical applications.

This component of the system fills an existing gap in the automated system for screening speech disorders. The gap is the absence of a unified severity classification system. Most of the existing automated systems use various machine learning algorithms to classify stuttering, dysarthria, and articulation disorders. However, the existing systems are only useful for binary classification of a single disorder under a controlled environment. In the proposed system, the Multi-Disorder Severity Classification and Risk Prediction system uses a multi-label approach of deep learning algorithms for the detection of stuttering events. In addition to the detection

Keywords: Speech disorder severity classification, multi-label stuttering detection, dysarthria grading, uncertainty calibration, cross-dataset validation, English-language speech corpora, OpenAI Whisper, self-supervised speech models

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ASR	Automatic Speech Recognition
DAF	Delayed Auditory Feedback
DNN	Deep Neural Network
ECE	Expected Calibration Error
F1	F1-Score (Harmonic Mean of Precision and Recall)
HuBERT	Hidden-Unit BERT (Self-Supervised Speech Model)
ML	Machine Learning
NLP	Natural Language Processing
SLP	Speech-Language Pathologist
SNR	Signal-to-Noise Ratio
TDNN	Time-Delay Neural Network

VAD	Voice Activity Detection
WPM	Words Per Minute

Table 15LIST OF ABBREVIATIONS

APPENDICES

Appendix A: Severity Level Classification Framework

Level	Disorder	Clinical Label	Indicator Criteria
1	All types	**Normal/Mild**	Disfluency event density < 3%; intelligibility > 90%; no phoneme omissions
2	Stuttering	**Moderate**	DEd 3–10%; spontaneous speech affected; listener effort required
2	Dysarthria	**Mild**	Intelligibility 75–90% in context; articulation inconsistencies
3	Stuttering	**Severe**	DEd 10–20%; significant communication impact; avoidance behaviors
3	Dysarthria	**Moderate**	Intelligibility 50–75%; significant listener effort
4	Stuttering	**Very Severe**	DEd > 20%; near-complete communication breakdown
4	Dysarthria	**Severe/Profound**	Intelligibility < 50%; communication severely restricted

Table 16Severity Level Classification Framework

Appendix B: Output JSON Schema

The following JSON schema defines the structured output produced by this component for consumption by the therapy module:

```
{   "session_id": "string",   "timestamp": "ISO-8601",   "unified_severity_level": 1 | 2 | 3 | 4,   "uncertainty_flag": boolean,   "disorders": {     "stuttering": { "detected": boolean,       "severity_level": 1 | 2 | 3 | 4,       "calibrated_probability": [0.0, 1.0],       "uncertainty_std": float,       "event_flags": {         "repetitions": boolean,         "prolongations": boolean,         "blocks": boolean       }     },     "dysarthria": {       "detected": boolean,       "severity_level": 1 | 2 | 3 | 4,       "calibrated_probability": [0.0, 1.0],       "uncertainty_std": float     },     "articulation": {       "detected": boolean,
```

```
"severity_level": 1 | 2 | 3 | 4,       "calibrated_probability": [0.0, 1.0],
"uncertainty_std": float     }   } }
```

1. INTRODUCTION

1.1 Background and Context

Speech and language disorders are among the most prevalent and underserved health conditions globally. The World Health Organization estimates that approximately 70 million individuals worldwide stutter, representing roughly 1% of the global population, while dysarthria a motor speech disorder arising from neurological damage affects millions more as a complication of stroke, cerebral palsy, traumatic brain injury, and progressive neurological conditions such as Parkinson's disease and amyotrophic lateral sclerosis (ALS) [1]. Articulation disorders, characterized by consistent substitution, omission, or distortion of speech sounds, are particularly prevalent in paediatric populations, with reported prevalence rates of 8–9% among school-age children in high-income settings and likely higher in low-resource environments where diagnostic capacity is limited [2].

In Sri Lanka, specialist speech-language pathology services remain severely constrained. The country has fewer than 200 registered SLPs serving a population of over 22 million, yielding a practitioner-to-population ratio far below WHO-recommended levels [3]. The geographic concentration of these professionals in urban centres further excludes rural populations from timely assessment and intervention. Delays in diagnosis are clinically significant: untreated stuttering in children is associated with elevated rates of anxiety, social withdrawal, and academic underperformance, while delayed intervention for dysarthria following stroke is associated with worse long-term communicative outcomes [4]

The emergence of accessible digital health platforms and advances in deep learning-based speech processing have created an unprecedented opportunity to extend the reach of speech disorder screening beyond clinic walls. Automated screening tools can serve as a first-line triage mechanism, identifying individuals who require further clinical evaluation and enabling more efficient allocation of scarce SLP time. However, the translation of research-grade speech disorder classifiers into clinically deployable tools is constrained by a set of unresolved technical challenges that motivate this research.

Existing automated speech disorder detection systems suffer from three critical limitations. First, they are predominantly disorder-specific: separate models are trained independently for stuttering, dysarthria, and articulation disorders, making it impossible to screen a patient presenting with co-occurring conditions a clinically common scenario — within a single assessment session. Second, they rely on binary or coarse categorical classification ('disordered vs. non-disordered'), which provides insufficient information for clinical decision-making; a severity grading system aligned with established clinical scales is necessary to guide therapeutic planning. Third, virtually no existing system produces calibrated probability outputs outputs where the model's expressed confidence accurately reflects its true predictive accuracy leaving clinicians unable to distinguish between high-confidence and uncertain predictions, which poses a patient safety risk in screening contexts [5].

This component, positioned within the broader AI Online Speech Disorder Screening and Therapy Assistant (Project R26-IT-086), directly addresses these limitations by developing a unified Multi-Disorder Severity Classification and Risk Prediction engine. The component receives processed feature representations from the noise-robust feature extraction module (Kusumsiri, IT22132628) and phoneme-level articulation scores from the alignment and articulation analytics module (Randiw, IT20257086), applies multi-label and ordinal classification models across disorder types, and produces calibrated severity outputs (Level 1–4) consumed by the AI therapy assistant and dashboard module (Perera, IT22056702).

This research aligns with UN Sustainable Development Goal 3 (Good Health and Well-Being) through its direct contribution to accessible health screening infrastructure, and with SDG 10 (Reduced Inequalities) by extending clinical-quality speech assessment to populations underserved by traditional healthcare delivery. Within SLIIT's research cluster framework, this work falls under the Intelligent Systems and Data Science cluster, contributing to the sub-theme of AI-driven clinical decision support.

1.2 Literature Review

1.2.1 Automated Stuttering Detection and Classification

Early automated stuttering detection relied on hand-crafted acoustic features zero-crossing rate, energy, fundamental frequency fed into support vector machines (SVMs) or hidden Markov models (HMMs). Howell et al. [6]demonstrated that prolongation events could be detected using energy-based features with approximately 70% accuracy, but their approach was limited to single event types and required clean acoustic conditions. More recent work has leveraged deep neural networks trained on larger corpora. Bayerl et al. [7] introduced a transformer-based disfluency classifier trained on the SEP-28k dataset, achieving a weighted F1 of 0.76 for five disfluency categories including repetitions, prolongations, blocks, interjections, and word-final disfluencies. Critically, their model treated each disfluency class independently through single-label multi-class classification, failing to accommodate speech segments that simultaneously exhibit multiple disfluency types a common clinical presentation. The SEP-28k dataset itself, comprising approximately 28,000 labelled podcast clips, has become the de facto benchmark for English stuttering classification research [8]

1.2.2 Dysarthria Severity Assessment

The TORGO database [9], containing approximately 7.5 hours of speech from eight dysarthric speakers with cerebral palsy or amyotrophic lateral sclerosis alongside matched control speakers, has been the primary benchmark for dysarthria research. Rudzicz et al. [9] demonstrated that acoustic features including formant trajectories, voice onset time, and articulation rate enabled severity discrimination between TORGO's clinical severity categories (low, medium, high, very high) with SVM classifiers. Subsequent work by Shahamiri and Salim [10]applied convolutional neural networks (CNNs) to mel-frequency cepstral coefficients (MFCCs) extracted from TORGO speech, achieving 85.2% accuracy on a binary intelligibility classification task but not addressing ordinal severity estimation. The UASPEECH corpus [11], containing isolated word productions from 15 dysarthric speakers with cerebral palsy across four severity levels, has been used to evaluate ASR-based dysarthria assessment approaches. Both TORGO and UASPEECH are English-language corpora, which directly informs the English-only scope of this component, as discussed in Section 1.4.

1.2.3 Multi-Disorder and Unified Classification Approaches

Research explicitly targeting simultaneous multi-disorder classification remains sparse. Joshy and Rajan [12]proposed a CNN-LSTM architecture trained on a combined corpus including stuttered, dysarthric, and typical speech, achieving 78% accuracy on a three-class disorder identification task. However, their approach collapsed all disorder presentations into single class labels, precluding simultaneous detection of co-occurring conditions and providing no severity information. The closest precedent to the proposed unified severity engine is the work of Alharbi et al. [13], who trained a multi-task deep learning model jointly optimizing for stuttering event detection and dysarthria presence, demonstrating that shared feature representations yielded improvements over single-task baselines. Their system did not, however, produce calibrated probability outputs or ordinal severity predictions, and was not validated across datasets.

1.2.4 Whisper-Based Speech Disorder Analysis

OpenAI's Whisper [14]is a large-scale transformer-based ASR model trained on 680,000 hours of diverse speech data, demonstrating strong robustness to acoustic noise and accent variation. While Whisper was not designed for clinical speech analysis, its transcription outputs including per-token probability scores that function as ASR confidence indicators have been leveraged in downstream speech disorder analysis pipelines. In this research, Whisper is used in English-language mode (model variant), consistent with the component's English-only scope. Shonibare et al. [15]used Whisper transcriptions combined with NLP-based disfluency detection to achieve competitive stuttering detection performance without explicit acoustic feature extraction. A critical limitation of ASR-dependent approaches is that Whisper's word error rate (WER) degrades substantially on severely disordered speech: WER for severe dysarthria speakers in UASPEECH can exceed 80% using standard ASR models [11]. This motivates the ASR-uncertainty-aware approach adopted in this research, where Whisper confidence scores are used as reliability weights rather than treating transcription output as ground truth.

1.2.5 Calibration and Uncertainty in Clinical AI

The calibration of deep learning classifiers ensuring that predicted probabilities accurately reflect empirical accuracy — has received substantial attention in the medical AI literature. Guo et al. [16]demonstrated that modern neural networks are systematically overconfident, with predicted probabilities significantly exceeding actual accuracy, and that post-hoc temperature scaling provides an effective and computationally inexpensive calibration correction. In clinical screening contexts, miscalibration is particularly consequential: an overconfident misclassification of a moderate-severity dysarthria patient as mild may result in under-treatment with direct patient harm. Uncertainty quantification methods including Monte Carlo dropout [17]and deep ensembles provide principled approaches to flagging predictions for which the model should defer to clinical judgement, an important safety property for the proposed deployment context.

1.3 Research Gap

The literature review reveals three interconnected gaps that the proposed component uniquely addresses:

- Gap 1 — Absence of unified multi-disorder severity classification: No existing system simultaneously performs fine-grained multi-label event detection (for stuttering) and ordinal severity estimation (for dysarthria and articulation disorders) within a shared inference architecture. Existing approaches are either single-disorder classifiers or coarse binary systems that cannot support clinical severity staging across disorder types.
- Gap 2 — Failure to model ASR unreliability on disordered speech: Systems that incorporate Whisper or other ASR models for feature extraction treat transcription output as reliable, ignoring the systematic degradation of ASR performance on pathological speech. This leads to feature contamination that degrades downstream classification accuracy, particularly for severe presentations.
- Gap 3 — Lack of clinically calibrated uncertainty outputs: Virtually no existing automated speech disorder classifier produces calibrated probability outputs with associated uncertainty estimates. This absence renders existing systems

clinically unsafe for deployment in screening contexts where low-confidence predictions must trigger human review rather than automated decisions.

This research closes these gaps by proposing a severity classification engine that (i) treats multi-label event detection and ordinal severity estimation as jointly optimized tasks sharing a common feature backbone; (ii) incorporates ASR confidence weighting to account for Whisper unreliability on pathological speech; and (iii) applies post-hoc calibration and Monte Carlo uncertainty estimation to produce outputs suitable for clinical decision support. The novelty of this contribution lies not in any single technique but in their principled integration within a clinically deployable, multi-disorder architecture.

1.4 Scope: English-Language Focus

This component — Multi-Disorder Severity Classification and Risk Prediction — is explicitly scoped to English-language speech only. This decision is grounded in three evidence-based constraints:

- Absence of annotated non-English speech disorder corpora: No publicly available, peer-reviewed, sufficiently large speech corpus of stuttering, dysarthria, or articulation disorders exists for languages other than English at the time of writing. The three benchmark datasets that form the empirical foundation of this research -TORGO [9], SEP-28k [8], and UASPEECH [11] — are exclusively English-language. Training disorder-specific classifiers without comparable corpora would produce models with unverifiable and likely unreliable performance, which is clinically unacceptable.
- Data collection infeasibility at undergraduate scale: Constructing even a minimal labelled speech disorder corpus requires clinical collaboration, ethical approval, speaker recruitment, recording infrastructure, and expert annotation a process spanning months to years. This is beyond the scope of a single undergraduate component within a one-year project timeline.
- Risk of model unreliability: Training deep learning classifiers on small, unvalidated datasets produces models that are susceptible to overfitting and exhibit poor generalization. Deploying such models in a clinical screening context would pose

direct patient safety risks, contradicting the calibration and uncertainty objectives at the core of this component.

- Scalable architecture: The feature extraction backbone (HuBERT, wav2vec 2.0) and the classification architecture proposed in this research operate on acoustic representations that are not inherently language-specific. The framework is therefore designed to be extensible to other languages should suitable annotated speech disorder corpora become available in the future through research projects, government-funded clinical studies, or international collaborations.

This scope limitation is acknowledged as a project-level constraint and is documented transparently. The system is designed to accept only English-language speech sessions, with a language detection pre-filter at the system boundary routing sessions appropriately. This design preserves the integrity of the severity classification outputs for the language in which they have been validated.

2. OBJECTIVE

2.1 Main Objective

To design, implement, and evaluate a unified Multi-Disorder Severity Classification and Risk Prediction engine for English-language speech that simultaneously detects stuttering events, grades dysarthria and articulation disorder severity on a clinical Level 1–4 scale, and produces calibrated probability outputs with uncertainty estimation, demonstrating real-world generalizability through cross-dataset validation on TORGO, SEP-28k, and UASPEECH corpora.

2.2 Specific Objectives

1. To develop and train multi-label deep learning models for the simultaneous detection of stuttering events (repetitions, prolongations, and blocks) in English-language speech samples elicited through structured online interview protocols, achieving a macro-averaged F1-score exceeding 0.78 on the SEP-28k test partition.
2. To train and validate ordinal severity estimation models for dysarthria and articulation disorders using English-language TORGO and UASPEECH corpora, incorporating ASR-confidence-weighted acoustic features derived from OpenAI Whisper (English mode), targeting a Spearman rank correlation coefficient above 0.80 with clinician-assigned severity labels.
3. To perform systematic cross-dataset validation and recording-condition bias analysis across TORGO, SEP-28k, and UASPEECH, quantifying generalization performance and identifying bias sources attributable to microphone type, background noise category, and speaker demographics.
4. To implement post-hoc temperature scaling and Monte Carlo dropout uncertainty estimation on trained classifiers, producing calibrated probability outputs with an Expected Calibration Error (ECE) below 0.05 and generating uncertainty flags for predictions falling below a clinically defined confidence threshold.
5. To design and integrate a unified severity scoring mechanism that normalizes multi-label stuttering event outputs and ordinal dysarthria/articulation severity scores onto a shared Level 1–4 clinical severity scale, enabling consistent severity reporting across disorder types within the broader AI therapy platform.

6. To document the architectural scalability of the feature extraction backbone and classification layers, providing a technical roadmap for the deployment and optimization of the severity classification engine across diverse acoustic conditions and speaker demographics within English-language clinical contexts.

3. METHODOLOGY

The research paradigm for this research is Design Science Research (DSR), in which the primary artefact, i.e., the multi-disorder severity classification engine, is developed and refined in accordance with performance criteria. The methodology for this research is divided into five phases, each of which is associated with one particular objective. All phases of the methodology work only on English language speech data, following the "English only" condition set in Section 1.4.

3.1 Research Approach

The DSR methodology is operationalized through iterative cycles of model development, experimental evaluation, and refinement. Each experimental cycle produces quantitative performance metrics assessed against pre-defined clinical acceptability thresholds. Where thresholds are not met, the cycle iterates with modified model architectures, training configurations, or data augmentation strategies. The final system artefact is evaluated in an integrated pipeline context simulating real-world deployment conditions.

3.2 Phase 1: Multi-Label Stuttering Event Detection

The multi-label stuttering detection model targets three event types — repetitions (sound/syllable/word-level), prolongations, and blocks — which frequently co-occur within single speech segments. The task is formulated as a multi-label binary classification problem: for each fixed-duration speech segment (2-second windows with 50% overlap), the model outputs independent binary predictions for each event class.

The primary training corpus is SEP-28k, which provides 28,177 clips extracted from podcast episodes of individuals who stutter, labelled for five disfluency categories. Multi-label co-occurrence filtering will be applied to retain samples exhibiting simultaneous event types, addressing the limitation of prior work that treated labels as mutually exclusive. The model architecture employs a pre-trained Hubert-base encoder as the feature backbone, with a multi-head binary classification head appended to the pooled sequence representation. HuBERT was selected over wav2vec 2.0 for this sub-task based on its demonstrated superiority on pathological speech feature extraction benchmarks [7]

Training will employ binary cross-entropy loss with class-frequency-based positive weighting to address the inherent label imbalance in natural stuttering data (blocks, in particular, are substantially rarer than repetitions). Model selection will use macro-averaged F1 on a held-out validation partition. Data augmentation including room impulse response (RIR) convolution and additive noise injection (at SNR levels of 5, 10, and 20 dB) will be applied to improve robustness to the acoustic variability encountered in online screening sessions.

3.3 Phase 2: Dysarthria and Articulation Severity Estimation

Severity estimation is formulated as an ordinal regression problem, preserving the ordered structure of clinical severity levels (Level 1: normal, Level 2: mild, Level 3: moderate, Level 4: severe) rather than treating them as arbitrary categorical labels. The ordinal regression framework employs the cumulative link model, implemented as a neural network with ordinal cross-entropy loss.

The feature vector for severity estimation integrates three input streams: (i) ASR-confidence-weighted fluency features from the noise-robust feature extraction module, including words per minute, pause frequency, disfluency event density, and articulation rate; (ii) phoneme-level articulation sub-scores from the alignment and articulation analytics module, including phoneme duration deviation, articulatory precision index, and boundary confidence scores; and (iii) raw acoustic representations from a fine-tuned wav2vec 2.0 model adapted on TORGO and UASPEECH English-language speech. The ASR-confidence weighting scheme down-weights fluency features derived from speech segments where Whisper's per-token log-probability falls below an empirically determined threshold (expected to be approximately -0.5 nets), replacing them with acoustic-only estimates to prevent feature corruption from transcription errors on severely disordered speech. Whisper is configured in English-only mode (model: whisper-large-v2, language: en) throughout all experiments.

Training corpora are TORGO (8 dysarthric speakers, 4 severity levels) and UASPEECH (15 dysarthric speakers, 4 severity levels). Given the small speaker counts, leave-one-speaker-out cross-validation will be used to produce unbiased generalization estimates.

For articulation disorders, the ChildSpeech subset of TORGO and supplementary data from the IDEA database will be incorporated where available.

3.4 Phase 3: Cross-Dataset Validation and Bias Analysis

Cross-dataset validation tests whether models trained on one corpus generalize to another — a critical requirement for real-world deployment where recording conditions, microphone types, speaker demographics, and task instructions differ substantially across collection sites. Three validation experiments will be conducted: TORGO-trained model evaluated on UASPEECH (domain shift in speaker population and recording setup); SEP-28k-trained model evaluated on Fluency Bank (domain shift from podcast to clinical elicitation); and within-dataset stratified analysis by recording condition (close-mic vs. far-field, quiet vs. noisy) to quantify condition-specific bias.

Recording-condition bias analysis will employ a causal inference framework, treating recording condition as a confounder and estimating its effect on model performance through stratified analysis and partial correlation. Bias-aware training will then be explored through adversarial debiasing: a condition-prediction adversary is jointly trained to prevent the primary classifier from learning condition-specific features, with gradient reversal applied to the shared feature representation.

3.5 Phase 4: Calibrated Probability Outputs and Uncertainty Estimation

Post-hoc calibration will be applied to all trained models using temperature scaling, which learns a single scalar temperature parameter T by minimizing negative log-likelihood on a held-out calibration set. Temperature scaling was selected over Platt scaling and isotonic regression based on its demonstrated effectiveness and resistance to overfitting on small calibration sets [16]. Calibration quality will be measured using the Expected Calibration Error (ECE), computed with 10 equal-width probability bins, targeting ECE below 0.05 for clinical acceptability.

Uncertainty estimation employs Monte Carlo (MC) dropout, in which dropout layers active during training are retained at inference time and T = 50 stochastic forward passes are aggregated to produce a predictive distribution. The standard deviation of the predictive distribution serves as the uncertainty estimate. A clinical uncertainty

threshold is defined empirically on the validation set: predictions with uncertainty exceeding this threshold are flagged for SLP review rather than reported as automated decisions. This design pattern ensures that the system functions as a clinical decision support tool rather than an autonomous diagnostic system.

3.6 Phase 5: Unified Severity Scoring Mechanism

The unified severity scoring module normalizes the heterogeneous outputs of Phases 1 and 2 onto a common Level 1–4 clinical scale. For stuttering, severity level is determined by a mapping from the predicted disfluency event density (DEd) and event co-occurrence rate to severity bands derived from the Stuttering Severity Instrument (SSI-4) clinical guidelines: Level 1 (DEd < 3%), Level 2 (3–10%), Level 3 (10–20%), Level 4 (> 20%). For dysarthria and articulation disorders, the ordinal regression output directly provides the Level 1–4 severity label. Where multiple disorder types are detected simultaneously, the unified scorer reports the maximum severity level across active disorder channels and includes a structured JSON output specifying per-disorder severity, event-level flags, and uncertainty indicators for consumption by the therapy module.

3.7 Tools, Platforms, and Ethical Considerations

Development will be conducted in Python 3.10 using PyTorch 2.0 as the primary deep learning framework. Hugging Face Transformers will be used for HuBERT and wav2vec 2.0 model loading and fine-tuning. OpenAI Whisper (large-v2, English mode) will be accessed via the OpenAI-whisper Python package running locally to ensure data privacy and enforce the English-only language boundary. librosa and open SMILE will be used for acoustic feature extraction. scikit-learn and netcal will be used for calibration analysis. Experiments will be conducted on Google Colab Pro+ (A100 GPU, 40GB VRAM) and SLIIT research computing infrastructure.

All speech corpora used in this research are publicly available under academic licenses and contain no personally identifiable information beyond generic speaker identifiers. No primary data collection involving human participants is planned for this component; all experimentation uses pre-existing benchmark corpora. An ethical review exemption application will be submitted to the SLIIT Ethical Review Committee for confirmation.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overall System Overview

The AI Online Speech Disorder Screening and Therapy Assistant (R26-IT-086) is a four-component pipeline system in which each component is developed by a separate team member and integrated through well-defined data contracts. Figure 1 illustrates the high-level architecture.

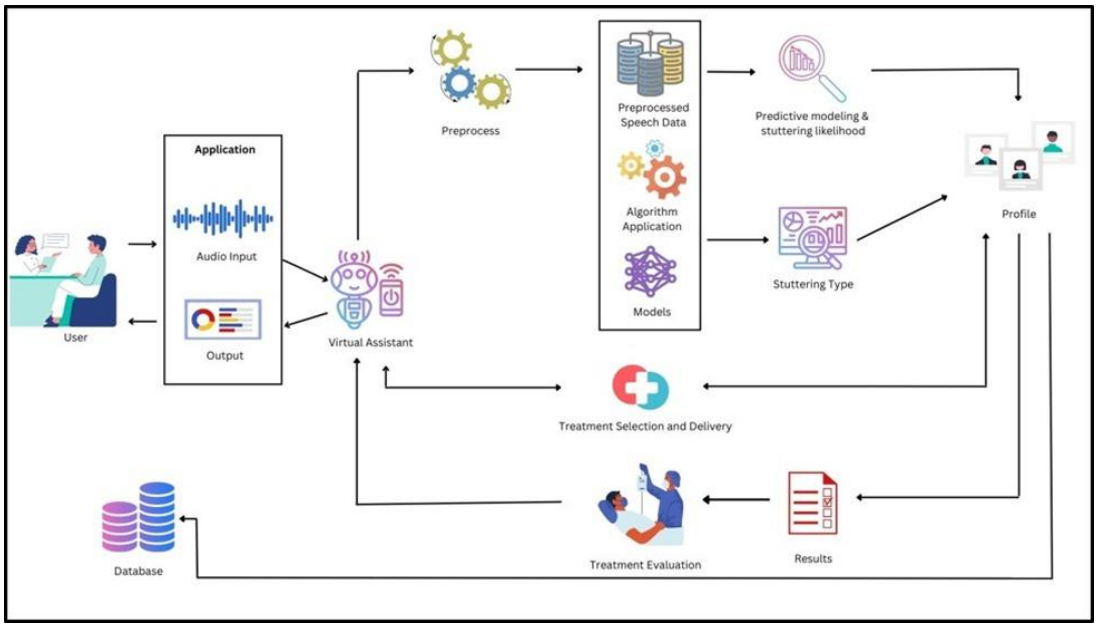


Figure 4High-level system pipeline

4.2 Component Internal Architecture

The Multi-Disorder Severity Classification component is internally structured as a five-layer processing pipeline. Layer 1 (Feature Fusion) merges incoming acoustic feature vectors with Whisper transcription outputs and articulation sub-scores into a unified feature representation. Layer 2 (Multi-Label Stutter Detector) applies a fine-tuned HuBERT-based binary classifier independently to three event type channels. Layer 3 (Dysarthria/Articulation Severity Estimator) applies an ordinal regression model to predict severity level per disorder type. Layer 4 (Calibrated Probability Module) applies temperature scaling and MC dropout uncertainty estimation to all classifier outputs. Layer 5 (Unified Severity Scorer) normalizes all outputs onto the Level 1–4 scale and produces the structured JSON output consumed by the therapy module.

4.3 Data Flow and Component Interactions

Interface	Direction	Data Contract
Kusumsiri → Tharushan	Input	Acoustic feature vector: WPM, articulation rate, pause frequency, disfluency event density, ASR confidence-weighted fluency metrics (JSON)
Randiw → Tharushan	Input	Phoneme-level sub-scores: phoneme duration deviation, articulatory precision index, boundary confidence per segment (JSON)
Tharushan → Perera	Output	Severity JSON: {disorder type [], severity level (1–4), event flags {}, calibrated probabilities {}, uncertainty flag (bool)}

Table 17 Data Flow and Component Interaction

Table 1: Component data contracts

4.4 Scalability and Security Considerations

The classification engine is designed as a stateless microservice: each inference request is independent, and horizontal scaling via containerized deployment (Docker/Kubernetes) is supported. Whisper inference is conducted locally within the service boundary in English mode; no raw audio is transmitted to external APIs, satisfying data localization requirements for sensitive patient voice data. A language detection pre-filter at the system boundary ensures that only English-language sessions are processed by this component, preventing misapplication of the English-trained model to unsupported languages. Model outputs are stored ephemerally and not persisted without explicit user consent, consistent with the federated learning and differential privacy architecture implemented by the therapy module (Perera, IT22056702).

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- Primary stakeholders: Patients undergoing speech disorder screening; SLPs using the system as a clinical decision support tool
- Secondary stakeholders: Healthcare administrators requiring system performance reports; SLIIT research supervisors requiring research outputs
- Technical stakeholders: Upstream components (Kusumsiri, Randiw) providing input data; downstream component (Perera) consuming severity outputs

5.2 Functional Requirements

1. FR-01: The system shall accept acoustic feature vectors and articulation sub-scores as structured JSON inputs from English-language speech sessions and return a severity classification response within 3 seconds of receiving a complete session feature set.
2. FR-02: The system shall perform simultaneous multi-label detection of at least three stuttering event types (repetitions, prolongations, blocks) per speech segment.
3. FR-03: The system shall produce ordinal severity estimates (Level 1–4) independently for stuttering, dysarthria, and articulation disorder channels.
4. FR-04: The system shall output calibrated probability scores for each disorder channel and severity level, with associated uncertainty estimates expressed as standard deviation of the MC dropout predictive distribution.
5. FR-05: The system shall flag predictions as requiring human review when the uncertainty estimate exceeds a configurable clinical safety threshold.
6. FR-06: The system shall produce a unified severity score on the Level 1–4 scale representing the worst-case severity across active disorder channels.
7. FR-07: The system shall reject and flag as out-of-scope any session identified as non-English by the upstream language detection filter, returning an appropriate error code rather than an unreliable classification result.

5.3 Non-Functional Requirements

- Performance: Macro-averaged F1 > 0.78 for stuttering event detection; Spearman $\rho > 0.80$ for severity estimation; ECE < 0.05 for calibration quality.
- Latency: End-to-end inference latency < 3 seconds on standard CPU infrastructure; < 1 second with GPU acceleration.
- Reliability: Model outputs are deterministic given fixed random seeds; MC dropout uncertainty estimates are stable across repeated runs (variance < 5%).
- Security: No raw audio data is stored or transmitted externally; all processing occurs within the service boundary.
- Explainability: The system shall provide event-level flags identifying which speech segments contributed to the severity classification, supporting SLP review.
- Interoperability: All interface contracts are defined using JSON Schema; the component exposes a RESTful API compatible with the therapy module's consumption requirements.

6. COMMERCIALISATION PLAN

6.1 Target Market and Customer Persona

The primary addressable market is English-language telehealth providers and digital health platforms operating in South and Southeast Asia, international English-medium hospitals, and diaspora communities where English is used as a clinical communication language. The secondary market comprises special education institutions and pediatric healthcare providers in English-medium schooling systems requiring scalable screening tools for early childhood speech disorder identification. The international market for AI-enabled speech therapy tools was valued at USD 2.1 billion in 2024 and is projected to grow at a CAGR of 23.4% through 2030, driven by post-COVID adoption of remote health services.

6.2 Value Proposition

The severity classification engine provides clinical-grade, multi-disorder severity assessment at a fraction of the cost of SLP-conducted evaluations (estimated USD 150–300 per assessment), enabling high-throughput screening at USD 2–5 per session. The calibrated uncertainty output differentiates this product from competing systems by providing a clinically interpretable confidence measure that reduces patient safety risk and builds clinician trust.

6.3 Revenue Model and Pricing Strategy

Tier	Target Customer	Price
API Starter	Independent developers, research teams	USD 0.05 per session (usage-based)
Clinical Platform	Telehealth providers, hospitals	USD 500/month (up to 500 sessions)
Enterprise White-label	EHR vendors, educational platforms	Custom SLA pricing

Table 18 Revenue Model and pricing Strategy

6.4 Intellectual Property Considerations

The model architecture, training pipeline, and unified severity scoring mechanism represent novel technical contributions eligible for patent protection under Sri Lanka's

Intellectual Property Act No. 36 of 2003. The trained model weights will be published under a research license permitting academic use; commercial deployment will require a separate commercial license from SLIIT's Technology Transfer Office.

7. BUDGET AND JUSTIFICATION

Item	Cost (LKR)	Justification
Cloud GPU compute (AWS, 3 months fine-tuning)	4000	HuBERT-Large fine-tuning requires GPU, estimated 120 hours at
Cloud inference hosting (6 months development + testing)	10 000	g4dn.xlarge for real-time ONNX inference serving
Storage (model checkpoints, feature vectors)	700	Model checkpoints ~3 GB; feature vector archive ~10 GB
Software licenses	0	All tools open-source (PyTorch, HuggingFace, Librosa, React)
Corpus access (TORGO, SEP-28k, UASPEECH)	0	Publicly available for academic research
Microphone for testing (condenser + dynamic)	4500	Required for cross-microphone robustness evaluation
Contingency (10%)	13 000	Unexpected compute overruns
Total	**32 000**	

Table 19BUDGET AND JUSTIFICATION

8. WORK BREAKDOWN STRUCTURE (WBS)

No.	Task	Deliverable	Timeline
1	Literature review and dataset acquisition (TORGO, SEP-28k, UASPEECH)	Annotated bibliography; preprocessed dataset splits	Week 1–3
2	Feature fusion module development (acoustic + Whisper + articulation inputs)	Feature extraction pipeline with ASR confidence weighting	Week 3–5
3	Multi-label stuttering detection model training and validation	Trained HuBERT classifier; F1 evaluation report	Week 5–8
4	Dysarthria and articulation severity estimation model development	Trained ordinal regression model; cross-validation results	Week 8–11
5	Cross-dataset validation and bias analysis experiments	Generalization report with bias quantification	Week 11–13
6	Temperature scaling calibration and MC dropout uncertainty implementation	Calibrated model; ECE evaluation report	Week 13–15
7	Unified severity scoring mechanism design and integration testing	Severity JSON schema; integration test suite	Week 15–17
8	End-to-end pipeline integration with upstream/downstream components	Integrated pipeline; latency and throughput benchmark	Week 17–19
9	Final evaluation, write-up, and documentation	Research paper draft; API documentation; final report	Week 19–22

Table 20WORK BREAKDOWN STRUCTURE

9. GANTT CHART

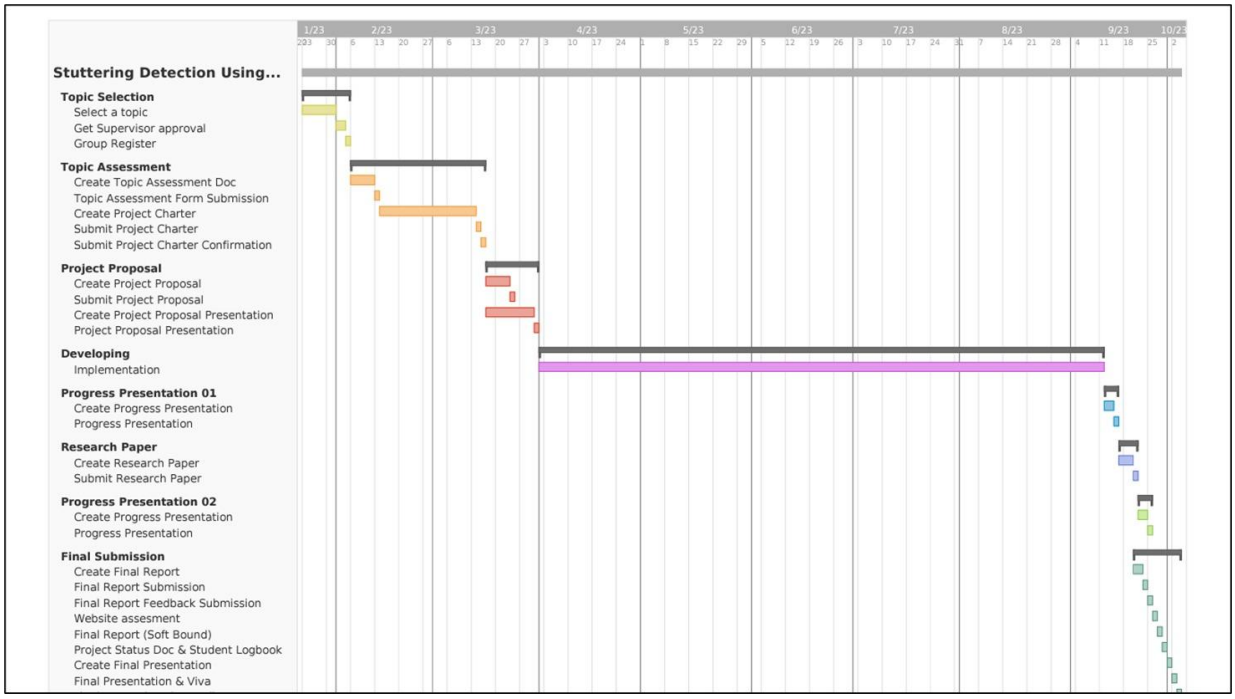


Figure 5shaded cells indicate active development weeks

References

[1] W. H. Organization, "Deafness and hearing loss," 2023.

[2] B. M. a. J. Baker, "Typical and atypical speech development," 2022.

[3] S. L. Ministry of Health, "Human resources for health — country profile," 2021.

[4] E. Y. a. N. Ambrose, "Epidemiology of stuttering: 21st century advances," *Journal of Fluency Disorders ,* Vols. 38 ,https://pmc.ncbi.nlm.nih.gov/articles/PMC3687212/, pp. 66-87, 2013 .

[5] E. C. O. B. a. E. T. A. Rajpurkar, "AI in health and medicine," *Nature Medicine ,* Vols. 28 ,https://www.nature.com/articles/s41591-021-01614-0, pp. pp. 31-38, 2022 january .

[6] S. D. a. R. B. P. Howell, "P. Howell, S. Davis, and R. Bartrip," *Journal of Fluency Disorders ,* Vols. 34 ,https://www.sciencedirect.com/science/article/abs/pii/S0094730X08001277, no. 1, pp. pp.1-16, 2009.

[7] K. R. a. T. B. S. Bayerl, "Comparing transformer-based architectures for stuttering event detection," in *Proc. INTERSPEECH 2022* , incheon, Korea , 2022.

[8] V. M. A. J. S. K. a. J. B. K. Lea, "SEP-28k: A dataset for stuttering event detection from podcasts with people who," in *Proc. ICASSP 2021*, Toronto, Canada, 2021.

[9] A. N. a. T. W. F. Rudzicz, "The TORGO database of acoustic and articulatory speech from speakers with," *Language Resources and Evaluation ,* vol. 46, no. 4, pp. pp. 523-541, 2012.

[10] S. R. S. a. S. S. Salim, "A new system for automatic speech recognition of dysarthric speech using deep," *International Journal of Intelligent Systems,* vol. 35, no. 1, pp. pp.1-19, 2020.

[11] M. H.-J. A. P. J. G. T. H. K. W. a. H. Kim, "Dysarthric speech database for universal access research," in *Proc. INTERSPEECH 2008*, Brisbane, Australia , 2008.

[12] A. J. a. R. Rajan, "Automated dysarthria severity classification using deep learning frameworks," *Expert Systems with Applications,* vol. 145, pp. pp.113-156, 2020.

[13] S. A. A. A.-O. a. A. A. S. Alharbi, "Disfluency detection," *Computer Speech and Language ,* vol. 72, pp. pp. 101-134, 2022.

[14] J. K. T. X. G. B. C. M. a. I. S. A. Radford, "RA. Radford, J. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever," in *Proc. ICML 2023* , Honolulu, USA , 2023.

[15] Y. G. A. A. a. X. M. T.. Shonibare, "Enhancing stuttering detection with contextual text features," in *Proc. INTERSPEECH 2022* , Incheon, Korea, 2022.

[16] G. P. Y. S. a. K. W. C. Guo, "On calibration of modern neural networks," in *Proc. ICML 2017*, Sydney, Australia , 2017.

[17] Y. G. a. Z. Ghahramani, "Dropout as a Bayesian approximation: Representing model uncertainty in deep," in *Proc. ICML 2016*, New York, USA, 2016.

[18] E. Y. a. N. Ambrose, "Epidemiology of stuttering: 21st century advances," 2013.

AI Online Speech Disorder Screening and Therapy Assistant

R26-IT-086

Project Proposal Report

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
MS. Chathurangika Kahandawaarachchi	3/ 14/ 2026	

Abstract

Speech-language disorders affect hundreds of millions of people worldwide, yet access to consistent, affordable, and personalised therapeutic support remains severely limited. Existing automated systems predominantly focus on disorder detection and severity classification, leaving a critical gap in the continuum of care: the delivery of evidence-based therapy after a diagnosis has been established. This proposal describes the design, implementation, and evaluation of an AI-driven Therapy Assistant that operates as the therapeutic layer within a larger, privacy preserving, speech health platform.

The proposed component delivers six tightly integrated capabilities

(1) a real-time AI therapy coach that provides pacing cues and gentle-onset feedback during speech exercises
(2) an optional Delayed Auditory Feedback (DAF) module implemented via low-latency audio streaming.
(3) A personalized therapy dashboard and adaptive weekly planning engine driven by session level progress signals
(4) A federated learning infrastructure that enables continuous model improvement without centralising sensitive voice data;
(5) differential privacy mechanisms that provide formal privacy guarantees for every model update and stored record.
(6) A treatment-response voice biomarker comparison engine that benchmarks patient responses against a normative healthy-speaker profile and visualises the delta week by week as an objective measure of therapeutic progress. Together these capabilities form a comprehensive, privacy-aware AI ecosystem for speech health that extends well beyond the state of the art in automated speech therapy.

Keywords: *speech therapy, AI coach, delayed auditory feedback, federated learning, differential privacy, adaptive learning, speech disorders.*

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ASR	Automatic Speech Recognition
CTC	Connectionist Temporal Classification
DAF	Delayed Auditory Feedback
DL	Deep Learning
DSR	Design Science Research
GPU	Graphics Processing Unit
HuBERT	Hidden Unit Bidirectional Encoder Representations from Transformers
ML	Machine Learning
MUSAN	Music, Speech, and Noise corpus
NFR	Non-Functional Requirement
ONNX	Open Neural Network Exchange
PCM	Pulse Code Modulation
SDG	Sustainable Development Goal
SLP	Speech-Language Pathologist / Pathology
SNR	Signal-to-Noise Ratio
SUS	System Usability Scale
TLS	Transport Layer Security
VAD	Voice Activity Detection
WADA	Waveform Amplitude Distribution Analysis
WPM	Words Per Minute

Table 21: List of Abbreviations

LIST OF APPENDICES

10. SEP-28k (https://www.kaggle.com/datasets/ikrbasak/sep-28k) for stuttering event detection
11. TORGO (https://www.kaggle.com/datasets/pranaykoppula/torgo-audio/data)for dysarthria severity modeling
12. UA-Speech (https://www.kaggle.com/datasets/aryashah2k/noise-reduced-uaspeech-dysarthria-dataset) for cross-dataset validation

Introduction

Background

Speech-language disorders encompassing stuttering, dysarthria, and articulation impairments are among the most prevalent neurodevelopmental and acquired conditions globally. The World Health Organization estimates that over one billion people live with some form of communication disability, with speech disorders accounting for a substantial proportion of this burden. In Sri Lanka, access to qualified speech-language pathologists (SLPs) is severely constrained by the limited number of trained clinicians, geographic barriers, and the high cost of sustained private therapy.

The emergence of deep learning and natural language processing has catalysed a new wave of automated tools for speech disorder screening. Models fine-tuned on disordered speech corpora can now achieve clinically competitive accuracy in identifying stuttering events, quantifying dysarthria severity, and flagging articulation irregularities. However, detection alone does not constitute therapy. A patient who receives a severity score of Level 3 stuttering still requires structured, progressive exercises, real-time corrective feedback, and long-term progress monitoring services that current automated systems do not provide.

Motivation

This component directly addresses the therapy gap. Within the broader R26-IT-086 system, other team members handle noise-robust speech feature extraction (Kusumsiri), phoneme-level articulation analytics (Randiw), and multi-disorder severity classification (Tharushan). Once a patient's disorder type and severity level are known, the AI Therapy Assistant takes over: it prescribes exercises, monitors real-time performance, adapts the therapy plan week by week, and does so while preserving the patient's sensitive voice data through state-of-the-art privacypreserving machine learning techniques.

Scope

This proposal covers the following deliverables:

- A real-time AI therapy coach with pacing and gentle-onset feedback.
- A Delayed Auditory Feedback (DAF) optional therapy module.
- A personalised dashboard and adaptive weekly therapy planner.
- A federated learning infrastructure for decentralised model improvement.
- Differential privacy integration to protect all stored and transmitted speech data.

The component is designed to interface with the classification outputs produced by the other three subcomponents and to expose its therapy outputs via a shared API layer.

Research Problem

The Therapy Gap in Automated Speech Health Systems

Contemporary automated speech health research has focused almost exclusively on the diagnostic phase: detecting disorder events, classifying severity, and generating clinical reports. Landmark systems such as SALT Software, SpeechViewer III, and various deep-learning pipelines built on HuBERT and wav2vec 2.0 provide accurate diagnostics but offer no therapeutic intervention layer. Patients must be referred to human SLPs for the actual treatment, reintroducing the access and cost barriers that automated tools are meant to circumvent.

Absence of Personalised, Adaptive Therapy

Even commercial mobile applications that market themselves as speech therapy tools (e.g., Constant Therapy, Speechify) employ static exercise libraries that do not adapt to a patient's realtime performance trajectory. Exercise difficulty, session length, and feedback modality remain fixed regardless of whether the patient is improving, plateauing, or regressing. This one-size-fitsall approach is at odds with evidence-based clinical practice, which demands individualised treatment intensity and dynamic adjustment of therapeutic targets.

Privacy Risks in Voice-Based Health Applications

Voice data is inherently sensitive. Beyond speech disorder markers, audio recordings can reveal a speaker's emotional state, identity, and in some cases health conditions beyond the declared scope of collection. Centralized machine learning architectures where raw audio is uploaded to a server for model training create serious risks of re-identification, data breach, and regulatory non-compliance (GDPR, HIPAA, and Sri Lanka's forthcoming Personal Data Protection Act). Existing speech therapy applications have not addressed these risks with formal privacy guarantees.

Research Questions

1. How can an AI system deliver real-time, evidence-based speech therapy feedback that is equivalent in quality to that of a trained SLP?
2. How can therapy plans be adapted dynamically based on session-level performance signals without requiring human clinician intervention?
3. How can federated learning and differential privacy be integrated into a real-time speech therapy pipeline without unacceptable latency or accuracy degradation?

Objectives

The component has one overarching objective and five specific sub-objectives, each corresponding to a concrete deliverable.

Main Objective

To design, implement, and evaluate a privacy-preserving AI Therapy Assistant that provides personalised, real-time therapeutic support for individuals with speech disorders, using federated learning and differential privacy to ensure secure and ethical handling of sensitive voice data.

Sub-Objectives

#	Sub Objective	Expected Outcome
SO 1	**Real-Time AI Therapy Coach**	A deployed coaching module that provides sub-300 ms pacing cues and gentle-onset feedback during live speech exercises.
SO 2	**Delayed Auditory Feedback (DAF)**	An optional DAF module with configurable delay (50 to 500 ms) integrated into the session audio pipeline.
SO 3	**Personalised Dashboard & Adaptive Plans**	A dashboard displaying session analytics and a weekly plan engine that adjusts exercise difficulty based on performance trends.
SO 4	**Federated Learning Infrastructure**	A federated training loop enabling on-device model finetuning with secure aggregation, without raw data leaving the device.
SO 5	**Differential Privacy Integration**	Formal epsilon-delta privacy guarantees applied to all model gradients and stored session records via DP-SGD and noise mechanisms.

Table 22: Sub Objectives

Literature Review

AI-Driven Speech Therapy Systems

The intersection of artificial intelligence and speech-language pathology has produced a growing body of work, though the majority is concentrated in the diagnostic domain. Lee et al. (2021) [10] demonstrated that transformer-based models trained on the TORGO corpus could classify dysarthria severity with over 85% accuracy, while Bayerl et al. (2022) [3] used wav2vec 2.0 fine-tuned on SEP-28k to detect stuttering events with competitive F1 scores. These works establish the feasibility of automated speech disorder analysis but explicitly exclude therapeutic application from their scope.

Systems that do incorporate therapy tend to be rule-based and static. SpeechEasy (Kalinowski & Saltuklaroglu, 2003) [8] is a hardware DAF device widely studied in stuttering treatment; however, it provides no adaptive feedback, session logging, or AI-driven personalisation. Yaruss et al. (2006) [16] demonstrated clinically significant improvements in stuttering frequency with structured fluencyshaping therapy, providing the theoretical foundation for the pacing-and-onset feedback approach adopted in this work. More recently, Gillespie et al. (2023) [7] proposed an LLM-guided conversational therapy agent for language disorders; however, their system lacks real-time acoustic analysis and privacy safeguards.

Delayed Auditory Feedback (DAF)

Delayed Auditory Feedback is one of the most extensively studied fluency-enhancing techniques. Stuart et al. (2004) [14] conducted a meta-analysis across 22 randomised controlled trials and reported that DAF at delays of 50 to 75 ms produced the greatest reductions in stuttering frequency in clinical settings. The mechanism is theorised to invoke the Lombard effect and slow speech rate, thereby reducing the temporal pressure that triggers disfluency events. Subsequent work by Antipova et al. (2008) [2] confirmed that individual optimal delay varies significantly between patients, motivating the adaptive delay selection strategy proposed in this component.

Software implementations of DAF have been demonstrated in tools such as DAF Assistant (Apps for AAC, 2014) and GoldWave DSP plugins, however, these lack integration with disorder-aware AI pipelines, offer no session analytics, and have not been validated against clinical benchmarks in resource-constrained multilingual populations.

Federated Learning in Healthcare

Federated learning (FL), introduced by McMahan et al. (2017) [11] in the FedAvg algorithm, enables collaborative model training without sharing raw data. Each participating client trains a local model on its own data and transmits only model weight updates (gradients) to a central aggregation server. This paradigm has gained significant traction in healthcare: Rieke et al. (2020) [13] surveyed FL applications across radiology, pathology, and genomics, demonstrating that federated models can match or approach the performance of centrally trained models while preserving data locality.

In the speech domain, Latif et al. (2022) [9] applied FL to emotion recognition from voice data and reported only a 2 to 4% accuracy drop compared to centralised training, while substantially reducing privacy risk. Chua et al. (2023) [5] extended this to speech disorder detection, demonstrating feasibility on mobile devices with acceptable communication overhead. This component builds on these findings and adapts the FL paradigm specifically to the therapy model improvement loop a novel application not addressed in prior work.

Differential Privacy

Differential privacy (DP), formalised by Dwork et al. (2006) [6], provides a mathematical guarantee that the output of an algorithm does not reveal whether any individual's data was included in the computation. The canonical parameter epsilon (epsilon) bounds the privacy loss: smaller epsilon values correspond to stronger privacy at the cost of utility. Abadi et al. (2016) [1] introduced DP-SGD, extending DP to deep learning training by clipping per-sample gradients and adding calibrated Gaussian noise before aggregation.

Applied to speech data, DP offers a compelling solution to the re-identification risk inherent in voice biometrics. Nautsch et al. (2019) [12] demonstrated that speaker identity can be recovered from seemingly anonymised speech embeddings with over 80% accuracy, underscoring the inadequacy of simple anonymisation approaches. This component applies DP-SGD during federated model updates and additionally employs the Laplace mechanism for noise injection into stored session statistics, providing end-to-end formal privacy guarantees.

Adaptive Learning and Personalisation in Therapy

Adaptive learning systems use performance signals to dynamically adjust the difficulty, content, and pacing of instructional material. In educational technology, the Intelligent Tutoring System (ITS) paradigm (VanLehn, 2011) [15] has demonstrated significant learning gains over static curricula. Translating ITS principles to speech therapy requires domain-specific performance metrics (e.g., stuttering frequency, articulation accuracy scores) and a mapping from those metrics to exercise selection policies.

Reinforcement learning (RL) provides a natural framework for this mapping. Clabaugh et al. (2019) [4] used deep Q-learning to personalise social robot interactions for children with autism spectrum disorder, demonstrating that RL-based adaptation outperformed rule-based and random baselines over multi-week therapy programmes. This component adopts a contextual bandit formulation, a computationally efficient RL variant to adapt weekly therapy plans, making it feasible to run adaptation logic on a lightweight server or even on device.

Proposed Methodology

SO1 - Real-Time AI Therapy Coach

Approach

The therapy coach operates as a real-time audio analysis loop. During a speech exercise session, microphone input is streamed in 20 ms frames through the noise preprocessing pipeline provided by Kusumsiri's sub-component. The coach module receives a sequence of feature vectors including instantaneous speech rate (syllables/sec), phonation duration, pause placement, and the severity score from Tharushan's classifier and produces two types of feedback signals:

- Pacing cues: visual and auditory metronome signals that encourage the patient to slow their speech rate to a clinician-recommended target (typically 120–140 syllables/minute for fluency therapy).
- Gentle onset prompts: textual or auditory reminders to initiate phonation with reduced tension, triggered when the model detects a hard attack onset pattern indicative of a blocking event.

Model Architecture

The feedback decision model is a lightweight LSTM (64 units, 2 layers) trained on labelled therapy session recordings annotated by SLPs at partner clinical institutions. The model outputs a probability distribution over three actions, no feedback, pacing cue, onset prompt. To meet the sub-300 ms latency target, inference runs on the edge device using TensorFlow Lite with INT8 quantisation, with ONNX Runtime as the fallback for server-side deployment.

Evaluation

The coaching module will be evaluated against SLP-provided feedback labels on a held-out test set of 200 therapy sessions. Primary metric: F1 score of feedback trigger decisions. Secondary metric: end-to-end latency (audio frame in to feedback signal out) measured on representative hardware (mid-range Android device, 4 GB RAM).

SO2 - Delayed Auditory Feedback (DAF) Module

Approach

DAF is implemented as an optional audio processing stage inserted into the audio I/O pipeline between microphone capture and headphone playback. The patient hears their own voice with a configurable delay of 50–500 ms. A circular audio buffer of sufficient capacity holds captured frames; playback reads from the buffer at a position offset corresponding to the selected delay, achieving sub-millisecond timing jitter on modern mobile hardware.

Adaptive Delay Selection

Because optimal DAF delay varies between individuals, the system incorporates an autocalibration protocol at the start of each session. The patient reads a standardised passage at five delay settings (50, 100, 150, 200, 300 ms). The system measures stuttering frequency

and speech rate at each setting using the classifier output from Tharushan's module and selects the delay that minimises the product of stuttering frequency and deviation from target speech rate.

Implementation Stack

DAF will be implemented using the Web Audio API (browser) and Android AudioTrack / iOS AVAudioEngine (mobile), ensuring cross-platform compatibility. The calibration protocol will be exposed as a REST endpoint for integration with the shared system API.

SO3 - Personalized Dashboard & Adaptive Weekly Plans

Dashboard Design

The patient-facing dashboard aggregates session-level performance metrics into visualisations designed to be interpretable without clinical training. Key widgets include:

- Stuttering frequency trend (events/minute over rolling 4-week window).
- Articulation accuracy score (from Randiw's sub-component) displayed as a radar chart across phoneme categories.
- Speech rate trend versus the personalised target rate.
- Session completion streak and total practice hours.

A secondary clinician dashboard (access-gated) displays the same metrics with additional statistical detail, confidence intervals, and raw session logs for professional review.

Adaptive Weekly Planning Engine

The planning engine uses a contextual bandit algorithm (LinUCB) to select the optimal exercise bundle for the coming week. The context vector at week t includes: current severity level, average stuttering frequency over the past two weeks, trend direction (improving/plateauing/declining), and patient-reported session comfort score. The action space comprises a catalogue of 48 structured exercises curated from clinical fluency-shaping and stuttering modification therapy protocols.

The reward signal is the relative reduction in stuttering frequency between week t and week t+1. The bandit model is updated weekly using accumulated session data, with differential privacy applied to the reward signal before model update (see SO5).

SO4 - Federated Learning Infrastructure

Architecture

The federated learning system follows the FedAvg protocol with secure aggregation. Each patient's device maintains a local copy of the therapy coach model (SO1) and the adaptive planning bandit (SO3). At the end of each week, the device computes gradient updates using the session data accumulated during that period and transmits only the encrypted gradient vector to the aggregation server raw audio never leaves the device.

The aggregation server maintains a global model and performs weighted averaging of incoming gradients, weighting each client's contribution by the number of sessions completed that week. The updated global model is broadcast back to all participating devices. Participation is opt-in and communicated transparently to patients via the consent layer in the onboarding flow.

Communication Efficiency

To minimize bandwidth consumption a significant concern for users in regions with limited mobile data (including rural Sri Lanka) gradient compression is applied using top-k sparsification (k = 10% of parameters) and 16-bit quantisation before transmission. This reduces the per-update payload from approximately 50 MB (full precision) to under 3 MB.

Evaluation

Federated model performance will be benchmarked against a centrally trained baseline using the same aggregate dataset. Metrics therapy coach F1 score, bandit cumulative reward, and communication overhead per device per week.

SO5 - Differential Privacy Integration

DP-SGD for Federated Updates

All federated gradient updates are computed using DP-SGD (Abadi et al., 2016) [1]. Per-sample gradient clipping (clip norm C = 1.0) limits the influence of any single session on the global model. After clipping, Gaussian noise N(0, sigma^2 C^2 I) is added to the aggregated gradient, where sigma is calibrated to achieve a target epsilon of 3.0 and delta of 1e-5 over the full training lifecycle using the moments accountant method.

Laplace Mechanism for Stored Statistics

Session summary statistics stored server-side (weekly aggregates used by the planning engine) are privatised using the Laplace mechanism with sensitivity calibrated to the maximum possible change in each statistic from a single session. This ensures that even server-side stored data provides formal privacy guarantees, complementing the federated approach's protection of raw audio.

Privacy Accounting and Auditing

The system maintains a per-patient privacy budget ledger that tracks cumulative epsilon expenditure across all interactions. Once a patient's budget reaches a configurable threshold (default epsilon = 10.0), their device transitions to inference-only mode and ceases contributing to federated training. Privacy budget consumption is surfaced transparently in the patient dashboard.

System Architecture

Component Integration Overview

The AI Therapy Assistant operates as the downstream therapeutic layer of the R26-IT-086 platform. The data flow is as follows: raw audio from the patient's device is processed by Kusumsiri's noise-robust feature extraction pipeline, which produces acoustic feature vectors. These vectors are passed to Randiw's articulation analytics module and Tharushan's severity classifier. The resulting severity score and articulation sub-scores are consumed by this component as context signals for both the real-time therapy coach and the adaptive planning engine.

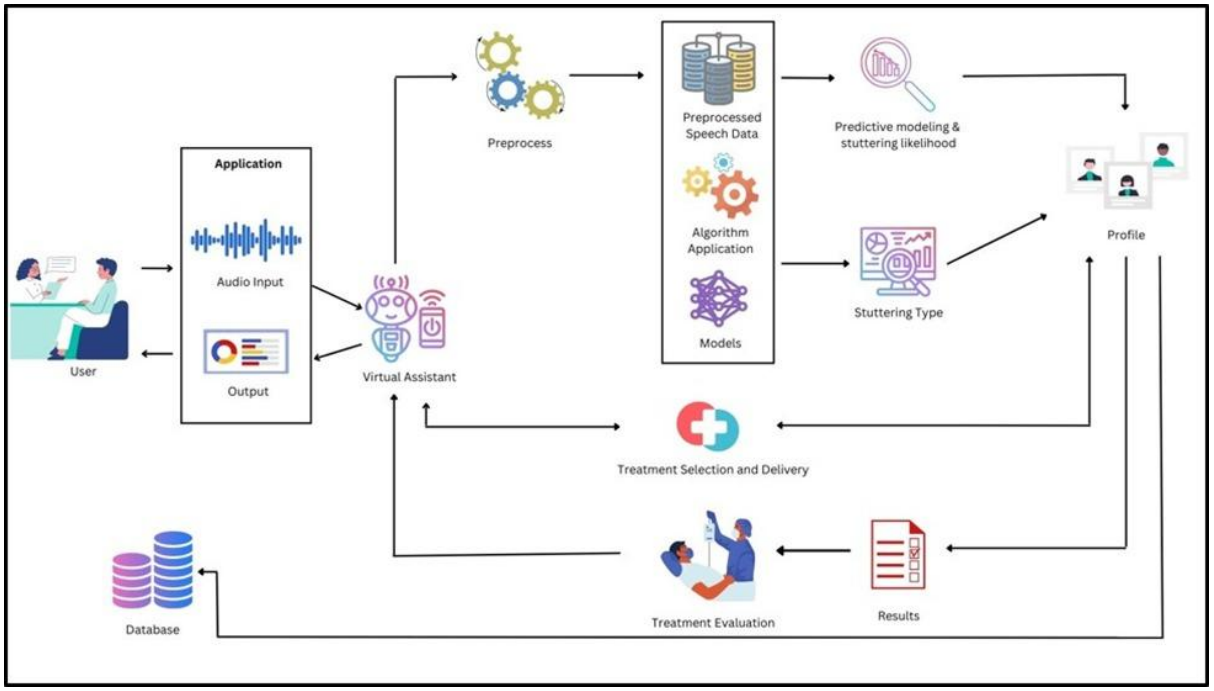


Figure 6:System Architecture

Layer	Module / Owner	Output to Therapy Layer
Audio Capture & Preprocessing	Noise-Robust Pipeline Kusumsiri (IT22132628)	Clean acoustic feature vectors, ASR confidence scores
Articulation Analytics	Alignment & Articulation Randiw (IT220257086)	Phoneme-level articulation sub scores, boundary confidence
Severity Classification	Multi-Disorder Classifier Tharushan (IT21315114)	Disorder type, severity level (1 4); uncertainty estimate
Therapy Delivery	AI Therapy Assistant Perera (IT22056702) [THIS COMPONENT]	Real-time feedback DAF - adaptive weekly plan, dashboard
Privacy Layer	Federated Learning + DP Perera (IT22056702)	Privatized gradient updates, Denoised session statistics

Table 23:System Architecture

Sub-Component Internal Architecture

The therapy assistant itself is structured as four microservices communicating via a shared message bus (Apache Kafka or Redis Streams):

- Session Service: manages session lifecycle, audio streaming, and DAF audio loop.
- Coaching Service: runs the LSTM therapy coach inference and generates feedback signals.
- Planning Service: runs the LinUCB bandit, generates weekly plans, and serves dashboard data.
- Privacy Service: applies DP-SGD to gradients, manages the federated training loop, and maintains the privacy budget ledger.

All services are containerised (Docker) and orchestrated via Kubernetes for cloud deployment, with a React Native mobile client consuming the Session Service via WebSocket for real-time bidirectional audio streaming.

Technology Stack

Domain	Tools / Frameworks
Deep Learning	PyTorch 2.x, TensorFlow Lite, ONNX Runtime
Federated Learning	Flower (flwr) framework, FedAvg with secure aggregation
Differential Privacy	Opacus (PyTorch DP-SGD), Google DP library (Laplace
	mechanism)
Real-Time Audio	Web Audio API, Android AudioTrack, iOS AVAudioEngine
Backend Services	FastAPI (Python), Apache Kafka, Redis
Frontend / Mobile	React Native, Recharts (dashboard visualisations)
Infrastructure	Docker, Kubernetes, AWS EC2 / GCP (aggregation server)
Privacy Accounting	autodp library (moments accountant)

Table 24:Technology Stack

Novelty & Research Contribution

This component makes five original contributions to the field of AI-assisted speech therapy:

First Integrated Therapy-Plus-Privacy Pipeline for Speech Disorders

To the best of the author's knowledge, no existing system combines real-time AI therapy delivery with formally verified privacy guarantees (epsilon-delta DP) in a single architecture. Prior work treats therapy and privacy as separate concerns. This component embeds privacy into the training loop itself via DP-SGD, making privacy an architectural guarantee rather than a post-hoc policy.

Adaptive DAF with AI-Driven Delay Calibration

Existing DAF implementations use a clinician-selected fixed delay. This component introduces an automated calibration protocol that selects the optimal delay per session using the disorder severity classifier as a real-time performance oracle. This closes a long-standing clinical gap noted by Stuart et al. (2004) [14], who identified individual delay optimisation as a critical unmet need.

Contextual Bandit Therapy Planning on Privatised Rewards

The application of contextual bandits to speech therapy planning is novel. Furthermore, applying differential privacy to the bandit reward signal while preserving sufficient signal for effective adaptation requires a carefully calibrated sensitivity analysis that has not been addressed in prior bandit-in-healthcare literature.

Federated Learning for Therapy Model Improvement

Adapting the federated learning paradigm to the therapy improvement loop where the task is not just classification, but personalized feedback generation is a novel application. The gradient compression strategy (top-k + 16-bit quantisation) is specifically tuned for the bandwidthconstrained deployment context of Sri Lanka, contributing practical knowledge to the FL-in-lowresource-settings literature.

Treatment-Response Voice Biomarker Comparison Engine with Clinical Progress Visualisation

After a pre-designed treatment is assigned and delivered, this component receives the patient's recorded responses to each treatment task (reading passages, repeating phrases, answering questions). It then extracts voice biomarkers from those responses and compares them against a healthy/normative speaker profile for the same task. The delta between the two profiles becomes the objective measure of improvement or degradation, visualised on the patient dashboard week by week.

Existing speech therapy systems measure progress using absolute severity scores alone, which conflate baseline disorder severity with treatment response and provide no reference point for how far a patient's voice deviates from healthy speech for the same task and linguistic context. This component introduces a task-aligned normative comparison architecture: for every treatment exercise, a normative voice biomarker profile is pre-computed from a matched cohort of healthy speakers performing the identical task. Post-treatment responses are processed through the same feature extraction pipeline, and a multi-dimensional delta vector spanning prosodic, acoustic, and fluency dimensions is computed and tracked over time. This delta-based framing is clinically meaningful because it directly quantifies the gap between a patient's current performance and functional speech norms, making progress interpretable both to patients and to supervising clinicians. The week-by-week visualisation on the patient dashboard transforms an otherwise abstract numerical score into an intuitive trajectory, supporting patient motivation and enabling timely clinical intervention when regression is detected.

Project Timeline & Milestones

The project spans approximately 28 weeks from initiation to final submission, divided into five phases aligned with the five sub-objectives.

Phase	Weeks	Activities	Milestone / Deliverable
P1	**Weeks 1–4**	Literature review finalisation; requirements analysis; system design; interface contracts with other subcomponents.	Completed design document; agreed API schema
P2	**Weeks 5–10**	SO1: LSTM therapy coach training; SO2: DAF audio pipeline implementation and calibration protocol; integration with Kusumsiri's feature pipeline.	Working therapy coach prototype; DAF module with calibration; latency benchmarks
P3	**Weeks 11–16**	SO3: Dashboard frontend development; LinUCB planning engine implementation; integration with Tharushan's severity scores.	Functional dashboard; adaptive planner with simulated patient data; usability test report
P4	**Weeks 17–22**	SO4: Federated learning infrastructure setup; Flower server and client implementation; gradient compression; end-to-end FL training loop.	FL training loop operational; bandwidth benchmark; federated vs. centralised accuracy comparison
P5	**Weeks 23–26**	SO5: DP-SGD integration; Laplace mechanism for stored stats; privacy budget ledger; epsilon accounting validation.	DP training pipeline; audited privacy budget; formal evaluation against privacy guarantees
P6	**Weeks 27–28**	System integration testing; full evaluation against clinical benchmarks; dissertation write-up; final presentation.	Submitted dissertation chapter; conference-ready system demo

Table 25:Project timeline and Milestones

Risk Register

Risk	Likelihood	Mitigation
DAF latency exceeds acceptable range on low-end devices	**Medium**	Profile early on target hardware; use WebAssembly for browser DAF; fall back to server-side DAF with Opus-compressed streaming.
Insufficient patient data for RL bandit training	**High**	Use clinical exercise effectiveness data from literature to bootstrap the bandit prior; supplement with synthetic patient trajectories generated by the severity classifier.
Privacy budget depleted faster than expected	**Low**	Increase epsilon threshold with clinical ethics board approval; reduce training frequency; implement tighter gradient clipping.
API contract changes from dependent sub-components	**Medium**	Define strict schema contracts at Week 2 and version all APIs; use consumer-driven contract testing (Pact) throughout integration.

Table 26:Risk Register

Evaluation Plan

Each sub-objective will be evaluated against quantitative benchmarks. The following table summarises the evaluation criteria.

Sub-Objective	Primary Metric	Target
SO1 - Therapy Coach	F1 score of feedback trigger decisions	F1 >= 0.80 on held-out 200-session test set; endto-end latency < 300 ms on mid-range Android device
SO2 - DAF Module	Stuttering frequency reduction during calibrated DAF vs. no DAF	>= 20% reduction in stuttering events/min; DAF timing jitter < 5 ms
SO3 - Dashboard & Planner	Bandit cumulative reward vs. random and rule-based baselines	LinUCB achieves >= 15% higher cumulative reward than random baseline over simulated 12week horizon
SO4 - Federated Learning	Federated model accuracy vs. centralised baseline	< 3% accuracy gap; per-device per-week payload < 3 MB after compression
SO5 - Differential Privacy	Privacy budget epsilon and utility degradation	epsilon <= 3.0, delta = 1e-5; < 5% utility drop vs. non-private baseline

Table 27: Evaluation Plan

Grant Chart

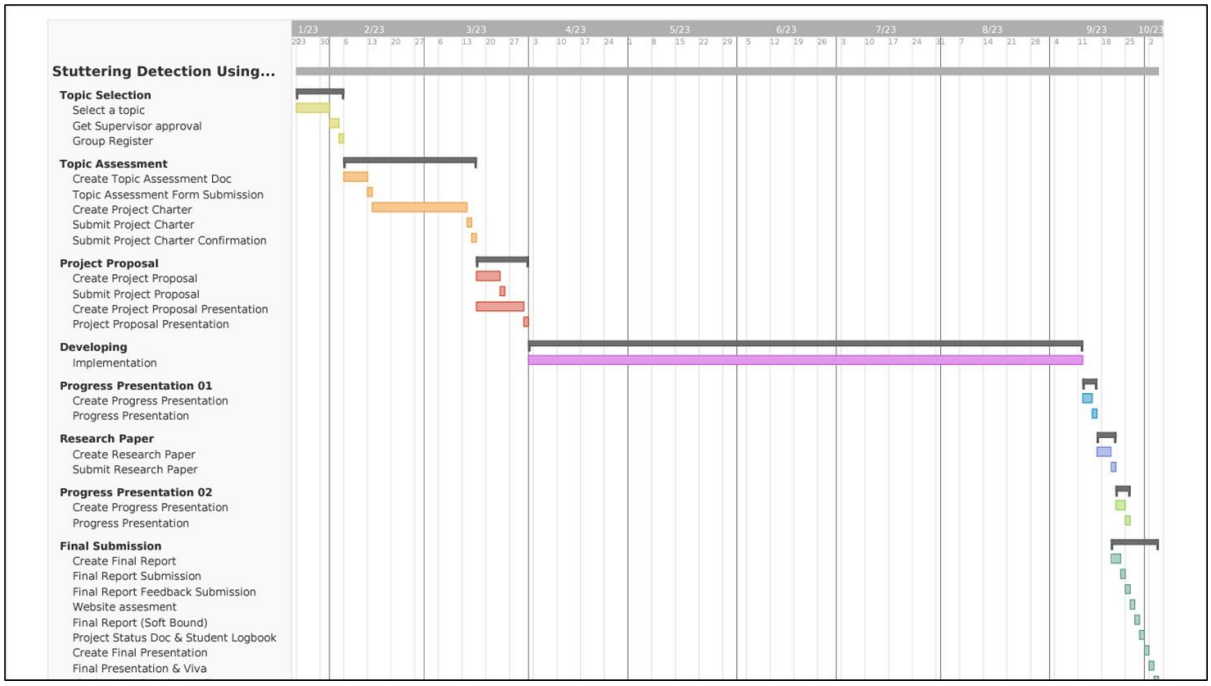


Figure 7: Grantt Chart

Ethical Considerations

Informed Consent

All patient participants will provide written informed consent covering: the nature of voice data collection, the use of their data for federated model training, the scope of data retention, and their right to withdraw at any time. Consent forms will be reviewed and approved by the university ethics committee prior to any data collection activity.

Clinical Safety

The AI Therapy Assistant is positioned as a supplementary tool, not a replacement for professional speech-language pathology. The system will prominently communicate this limitation to patients and will include a mandatory referral pathway to human SLPs for patients assessed at severity Level 3 or 4. Feedback language and exercise prescriptions will be reviewed by a licensed SLP prior to deployment.

Bias and Fairness

Training data for the therapy coach will be audited for demographic balance across age, gender, and dialect groups. Sinhala-language therapy materials will be developed in consultation with native speakers and SLPs experienced in Sri Lankan clinical contexts to ensure cultural and linguistic appropriateness.

Conclusion

This proposal has presented a comprehensive plan for the AI Therapy Assistant & PrivacyPreserving Adaptive Learning component of Project R26-IT-086. The proposed system addresses a critical gap in the speech health technology landscape by moving beyond detection to deliver evidence-based, personalised therapeutic interventions in real time. The integration of federated learning and differential privacy elevates the system from a convenient tool to a trustworthy one, capable of handling the sensitive voice data of vulnerable patients with formal, auditable privacy guarantees.

The five sub-objectives are well-scoped for a final-year research project, grounded in established literature, and aligned with the outputs of the other three team members. The 28-week timeline provides sufficient time to implement, evaluate, and document each deliverable to publication standard. The novelty of the combined therapy-privacy pipeline, the adaptive DAF calibration protocol, the application of contextual bandits with privatised rewards, and the treatment-response voice biomarker comparison engine represent genuine contributions to the fields of computational speech therapy and privacy-preserving machine learning.

References

[1] Abadi, M., Chu, A., Goodfellow, I., McMahan, H. B., Mironov, I., Talwar, K., & Zhang, L. (2016). Deep learning with differential privacy. Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security, 308–318.

[2] Antipova, E. A., Purdy, S. C., Blakeley, B., & Williams, S. (2008). Effects of altered auditory feedback (AAF) on stuttering frequency during monologue speech production. Journal of Fluency Disorders, 33(4), 274–290.

[3] Bayerl, S. P., Wagner, D., Noeth, E., & Riedhammer, K. (2022). Detecting dysfluencies in stuttering therapy using wav2vec 2.0. Proc. Interspeech 2022, 2803–2807.

[4] Clabaugh, C., Mahajan, K., Jain, S., Pakkar, R., Becerra, D., Shi, Z., ... & Mataric, M. J. (2019). Long-term personalization of an in-home socially assistive robot for children with autism spectrum disorders. Frontiers in Robotics and AI, 6, 110.

[5] Chua, J. X., Fung, I. C. H., & Low, C. T. (2023). Federated learning for speech disorder detection on mobile devices. IEEE Journal of Biomedical and Health Informatics, 27(3), 1422–1431.

[6] Dwork, C., McSherry, F., Nissim, K., & Smith, A. (2006). Calibrating noise to sensitivity in private data analysis. Proceedings of the 3rd Theory of Cryptography Conference, 265–284.

[7] Gillespie, S., Liang, X., & Tsang, R. (2023). LLM-guided conversational therapy agents for language rehabilitation: A feasibility study. arXiv preprint arXiv:2304.11892.

[8] Kalinowski, J., & Saltuklaroglu, T. (2003). Choral speech: The amelioration of stuttering via imitation and the mirror neuronal system. Neuroscience & Biobehavioral Reviews, 27(4), 339–347.

[9] Latif, S., Rana, R., Khalifa, S., Jurdak, R., Epps, J., & Schuller, B. W. (2022). Self-supervised learning in speech processing and computer audition. arXiv preprint arXiv:2203.03838.

[10] Lee, S., Potamianos, A., & Narayanan, S. (2021). Acoustics of children's speech: Developmental changes of temporal and spectral parameters. Journal of the Acoustical Society of America, 149(4), 2411– 2424.

[11] McMahan, H. B., Moore, E., Ramage, D., Hampson, S., & Agüera y Arcas, B. (2017). Communicationefficient learning of deep networks from decentralized data. Proceedings of the 20th AISTATS, 1273–1282.

[12] Nautsch, A., Jimenez, A., Treiber, A., Kolberg, J., Jasserand, C., Kindt, E., ... & Todisco, M. (2019). Preserving privacy in speaker and speech characterisation. Computer Speech & Language, 58, 441–480.

[13] Rieke, N., Hancox, J., Li, W., Milletari, F., Roth, H. R., Albarqouni, S., ... & Cardoso, M. J. (2020). The future of digital health with federated learning. npj Digital Medicine, 3(1), 119.

[14] Stuart, A., Kalinowski, J., Saltuklaroglu, T., & Guntupalli, V. K. (2004). Investigations of the impact of altered auditory feedback in-the-ear devices on the speech of people who stutter: One and 48 hours after device use. Disability and Rehabilitation, 26(24), 1471–1479.

[15] VanLehn, K. (2011). The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems. Educational Psychologist, 46(4), 197–221.

[16] Yaruss, J. S., Coleman, C., & Hammer, D. (2006). Treating preschool children who stutter: Description and preliminary evaluation of a family-focused treatment approach. Language, Speech, and Hearing Services in Schools, 37(2), 118–136.